

Negative-set construction moves measured model contribution more than five-fold, and the estimator returns a positive value where the truth is zero: a calibration across 94 ENCODE eCLIP datasets

Nirmalkumar Thirupallikrishnan Kesavan
Northeastern University, Boston, MA, USA
thirupallikrishnan.n@northeastern.edu

Abstract

Sequence models for RNA-binding proteins are evaluated against constructed negative windows, and how those windows are chosen is usually treated as a detail. Across 94 paired ENCODE eCLIP datasets we held the model, the peak set, the chromosome-blocked folds and the estimator fixed, varied only how the negatives were built, and measured each model’s nested contribution over a 19-feature composition baseline refit under each protocol. Replacing GC-matched with dinucleotide-matched negatives lowered the 4-mer’s own AUROC from a panel mean of 0.798 to 0.688 and raised its measured contribution from 0.027 to 0.066. The fall held in every one of the 94 datasets individually and the rise in 88 of them, not only in the panel means. The direction is implied by the design, since dinucleotide matching constrains fifteen degrees of freedom of composition against GC matching’s one, so only the magnitude is informative. A bias-aware protocol drawing negatives from other proteins’ binding sites left the highest composition baseline of the three and the smallest contribution, 0.012, and there nineteen composition features beat the 4-mer outright in 53 of 94 datasets. It is also the easiest discrimination for the 4-mer and the convolutional network, though not for SpliceBERT, which finds GC matching marginally easier. Across the three protocols apparent difficulty and measured contribution move in opposite directions, spanning 5.42-fold for a 4-mer and 3.7 to 7.4-fold across three model classes on identical rows. A fourth construction shows what the protocol is acting on. Dinucleotide-preserving shuffling makes each negative a permutation of its own positive, so the composition baseline takes the same value on both members of every pair and sits at exactly 0.5000 in all 94 datasets. The contribution there is +0.2523, which with the baseline pinned at a half is the model’s own AUROC less a half rather than an independent measurement. Across the four constructions the span is 20.62-fold. For a 4-mer what a negative-set protocol moves is the composition baseline, and the contribution follows; for the convolutional network the two attributions reverse. On the one externally constructed benchmark we tested they move together, so the relation is a property of the protocols compared here and not a general law. No standard way of re-expressing an AUROC difference closed the span. We tried eight: the raw difference, Somers’ D, discriminability, the logit, arcsine and complementary log-log scales, and division by the baseline’s remaining

headroom or by its excess over chance; five shrank the span but not below two-fold, two are controls that return it unchanged by construction, and one widened it. An exponent fitted so as to close it survived neither a change of aggregation nor transfer to an independent benchmark. At an order-three baseline most of the 4-mer’s contribution disappears and the span widens, so the quantity is indexed by the baseline’s order as well as the protocol. Headroom division improved cross-protocol agreement on our own data but failed, out of sample, a falsification test specified in advance. The estimator also has a floor. Applied to a model whose information the baseline already contains, so that the true contribution is zero by construction, it returns +0.0111 to +0.0137: nearly constant across protocols, so the span is not manufactured by it, but 90.4% of the smallest arm’s reported contribution, so absolute increments of this size are not distinguishable from it. In a survey of seven widely used methods and benchmarks, five construct negatives by relocating genomic intervals, which leaves composition unconstrained, none reports a composition-only AUROC beside its headline, and none reports a calibration that would reveal the floor. We therefore recommend reporting the composition-only AUROC obtained under the same protocol alongside every headline AUROC, and not comparing contributions measured under different protocols.

Keywords: RNA-binding proteins; eCLIP; benchmarking; negative-set construction; sequence composition; incremental value; AUROC

1 Introduction

Sequence models that predict RNA-binding protein (RBP) occupancy are evaluated almost entirely by the area under the receiver operating characteristic curve (AUROC) on held-out windows. Positive windows come from a crosslinking assay, most often eCLIP [2, 3], but negative windows have to be constructed, and the construction is usually given a sentence in the methods. Where it is specified, negatives are commonly unbound regions of the same transcripts [20] or a composition-matched genomic sample [6]. Horlacher *et al.* [4] benchmarked eleven RBP prediction methods across hundreds of CLIP-seq experiments under two negative-set definitions and showed that a method’s apparent performance depends on which definition is used. They proposed a bias-aware alternative in which negatives are drawn from the binding sites of other RBPs rather than from genomic background. The same dependence has since been reported for transcription factor binding sites, where five negative-sampling strategies were compared against a position weight matrix lower baseline and models trained on dinucleotide-shuffled negatives transferred poorly to held-out high-quality data, at about 76% AUROC against about 86% for genomic sampling [5].

How large that dependence is, and in which direction it acts, remains open. A raw AUROC conflates the signal a model found with the difficulty of the discrimination it was set, and whoever builds the benchmark controls the second. Negatives sampled uniformly from the genome differ from crosslinked positives in base composition, so a classifier responding to nucleotide frequency alone will score well without having learned anything sequence-specific. Matching negatives on GC content is the usual mitigation, but GC content is one linear functional of the sixteen-cell dinucleotide simplex, which has fifteen degrees of freedom, so fourteen of them remain free for any model to

exploit. Comparable circularity is characterised in variant-effect prediction, where the assembly of the evaluation set both inflates apparent accuracy and reorders methods [10], and the sensitivity of genomic machine-learning evaluations to the definition of the negative class is recognised more generally [7, 8, 11]. That a difference in AUROC between two models depends on how strong the baseline model is has also been characterised outside genomics [15].

We call a model’s own AUROC on a given protocol its *apparent* AUROC, since it is the number a benchmark reports and it depends on the protocol as much as on the model. A model’s increment over an explicit composition baseline is less exposed to compositional confounding than its apparent AUROC. We define the nested contribution as the difference between the pooled out-of-fold AUROC of a logistic model on composition features together with the model’s score, and that of the composition features alone, fitted on the same rows and the same folds. Every contribution reported here is therefore an increment over an order-two baseline: the mono- and dinucleotide frequencies plus sequence entropy, which is nineteen columns once one level is dropped from each frequency family to keep the design matrix non-singular. That choice is not innocuous, and the Discussion measures what a third order costs.

We assembled 94 ENCODE eCLIP datasets and constructed negatives three ways: matched on GC content, matched on all sixteen dinucleotide frequencies, and drawn from the binding sites of other RBPs assayed in the same cell line, following Horlacher et al. [4]. The three protocols differ in what they hold fixed, and not only in how tightly they match composition: the two composition-matched protocols also draw candidates of the same transcript region class, which the bias-aware protocol does not, and only the bias-aware protocol’s negatives are transcribed RNA. Model, peak set, chromosome-blocked folds and estimator were held fixed across arms, and the composition baseline was refit on each arm’s own windows so that no baseline was carried between designs. Three model classes were measured under all three protocols: a 4-mer logistic regression, a 7089-parameter convolutional network, and SpliceBERT [21], a fully fine-tuned 19.78 M-parameter nucleotide language model.

Apparent difficulty and measured contribution move in opposite directions. Dinucleotide matching reduced apparent AUROC in all 94 datasets while raising the nested contribution by a factor of 2.50 in panel means, and the bias-aware protocol left the highest composition baseline of the three and the smallest contribution. The contribution spanned 5.4-fold across protocols for the 4-mer, and between 3.7 and 7.4-fold across the three model classes.

For scale, the two comparisons a benchmark exists to support can be put in the same units. Changing protocol while holding the model fixed moves the panel-mean contribution by 0.0540 for the 4-mer, 0.0724 for the convolutional network and 0.1269 for SpliceBERT. Changing model class while holding the protocol fixed moves it by 0.0625 in the GC arm, 0.1073 in the dinucleotide arm and 0.0353 in the bias-aware arm. The two are comparable and neither dominates. In absolute AUROC, which is larger depends on the arm and the model one starts from. As a fold change the protocol range reaches higher but still overlaps the other, 3.7 to 7.4-fold across protocols against 2.6 to 4.1-fold across model classes, and SpliceBERT’s protocol span falls below the bias-aware arm’s

model-class span. The claim is not that the protocol matters more than the model. It is that it matters as much, and is usually not reported. Of eight standard coordinates five reduced the span without closing it, two are controls and one increased it, and an exponent fitted to close it survived
 115 neither a change of aggregation nor transport to an externally constructed benchmark. On that basis we recommend reporting the composition-only AUROC alongside every headline AUROC, and we report where that helps and where it does not.

2 Materials and Methods

2.1 Panel selection

120 Candidate experiments were every released ENCODE eCLIP experiment in K562 and HepG2 (139 and 105 respectively). Where a protein had several peak files we took the one with the most biological replicates, breaking ties by ENCODE’s `preferred_default` flag, and required `output_type=peaks`, `assembly=GRCh38`, `status=released` and `file_format=bed`. Every candidate was preprocessed under both composition-matching protocols; datasets retaining at least 400 out-of-fold scored pairs
 125 were eligible.

The study panel was then defined once by the dinucleotide arm, which is the stricter control: the 189 eligible datasets were sorted by pair count with a stable sort and every second one retained, giving 95 datasets over 79 proteins. The set was halved rather than taken whole because the sweep trains three model classes over five folds and three protocols, and the full eligible set did not fit the
 130 compute budget the retained panel already consumes. Given that a reduction was necessary, we sampled systematically by rank rather than taking the largest N . AUROC correlates with dataset size in these data: negligibly for the composition baseline (Pearson r on $\log_{10} n$ of -0.04 in the GC arm) but strongly for the models, reaching 0.70 for SpliceBERT under dinucleotide matching. A size-truncated panel would therefore confound panel membership with the measured quantity. The
 135 retained panel spans the 0th to 99th percentile of the candidate size distribution. Of the 95, 94 also cleared the floor in the GC arm and constitute the paired panel used for every three-protocol result, giving 282 protocol-dataset cells. The 95th carries no three-protocol result, since it did not clear the out-of-fold pair floor in the GC arm, and every analysis reported here is on the 94. Panel membership was written once to a manifest that every downstream stage reads. Supplementary
 140 Table S1 lists every dataset with its ENCFE file and ENCSR experiment accession.

Sixteen proteins are assayed in both cell lines and contribute two datasets each, fifteen of them in the paired panel. All headline intervals therefore resample proteins rather than datasets.

2.2 Windows

Positive windows are 101 nt centred on the midpoint of each peak interval. Peak width and the
 145 significance columns are unused: we applied no further p -value, q -value or fold-change filter, because ENCODE’s released `peaks` files are already thresholded and across our 95 files the minimum $\log_2$ fold-enrichment is 3.0000 and the minimum $-\log_{10} p$ is 3.0003, which are the criteria of Van Nostrand

et al. [3]; 91.7% of the 463,091 peaks carry ENCODE’s IDR label. The positive set is therefore every peak ENCODE called and no weaker ones. Minus-strand windows are reverse-complemented and transcribed to RNA; every sequence is the strand the protein sees. Windows containing an ambiguous base, falling outside the assembly, duplicating an existing window by start coordinate, or carrying no region annotation are dropped and counted. Analysis is restricted to chr1 to chr22 and chrX; chrY and all scaffolds and alternates are excluded.

Region class is assigned from GENCODE v45 [9] as the first of (5’UTR, 3’UTR, CDS, non-coding exon, intron) whose interval set contains the window midpoint. Region intervals were built from all transcripts with no gene-type filter and merged, with strand disregarded at this step; introns were computed as the gaps between consecutive exons. GTF 1-based inclusive coordinates are converted to 0-based half-open.

2.3 The three negative-set protocols

Each protocol produces one negative per positive. All three use a single random stream seeded at 7 per protein.

GC-matched. Candidates are windows of the same region class on the same chromosome, at least 500 nt from any peak of the target protein, drawn with probability proportional to the length of each free interval. A candidate is accepted if its GC content is within 0.05 of the positive’s. The matcher relaxes: after 25 unsuccessful draws the tolerance doubles to 0.10, and if 40 draws fail the best candidate seen is accepted provided it is within three times the nominal tolerance, that is 0.15. Only the target protein’s own peaks are excluded, and a GC-matched or dinucleotide-matched negative may be another protein’s binding site. The negative inherits the positive’s strand label; the consequences are quantified in the Results.

Dinucleotide-matched. Positives are grouped by region and chromosome, and for each group a candidate pool of $\max(1500, 8n)$ windows is sampled from the same free intervals the GC matcher draws on: same region class, same chromosome, and at least 500 nt from any peak of the target protein, the exclusion applied through the same code path and the same configured distance. The two composition-matched protocols therefore share one candidate universe and differ only in the statistic they match on. For each positive the nearest unused candidate is assigned greedily under the L_1 distance over the sixteen dinucleotide *counts*, searched with a k -d tree over the forty nearest neighbours. The distance is over counts, not frequencies, because integer distances are exactly representable, and each candidate carries an additional tiebreak column that increases strictly with genomic position and is bounded below 0.5, so it can never reorder candidates whose integer distances differ but resolves exact ties by position. Without both devices the choice among tied candidates depends on the floating-point behaviour of the k -d tree implementation and a different negative is selected for about 8% of pairs between CPU architectures. Assignment is greedy, not optimal; achieved distances are reported so the quality of the approximation is visible. Used coordinates are

tracked across all groups, because one interval can be annotated as both a non-coding exon and a
 185 3'UTR and so appear in two region pools. No maximum distance is imposed.

Bias-aware (other RBPs' sites). Following Horlacher et al. [4], negatives for a target are the
 positive windows of the other panel proteins assayed in the same cell line, sampled 1:1 without
 replacement, excluding any donor window within 500 nt of any of the target's own positive windows.
 Our version deviates from Horlacher et al. [4] in two ways: our donor pool is the study panel (40
 190 to 48 donor proteins per dataset, median 45) and not a full ENCODE survey, and exclusion is by
 distance from the target's 101 nt windows, not from its full peak intervals. Both the donor pool and
 the target's positives are read from the GC arm's window tables; this arm inherits the GC matcher's
 retained positive set and its pair count is identical to that arm's. Sampling is performed within fold,
 and every pair stays inside one cross-validation fold. These negatives are transcribed, accessible to
 195 crosslinking and strand-correct by construction, and are not composition-matched in any respect.

What each protocol holds fixed. The three protocols are not three settings of one knob,
 and the difference changes how their results should be read. Both composition-matched protocols
 draw candidates of the same transcript region class as the positive, so the region label carries no
 information about the class. The bias-aware protocol matches fold only, which leaves region free;
 200 how far region separates the two classes there is measured in Section 3.2. Only the bias-aware
 protocol's negatives are transcribed RNA, and conversely 67.6% of GC-arm negatives lie in loci
 not transcribed on the strand the window is labelled with. Section 3.2 bounds what the region
 asymmetry contributes; the strand and transcription controls in the Results bound the other two;
 these are the tables the repository names `strand_placebo` and `transcription_placebo`.

2.4 Achieved match quality

Achieved matching was quantified over every pair in each arm, 456,734 in the GC and bias-aware
 arms and 457,998 in the dinucleotide arm (Table 1).

Table 1: Achieved match quality between each positive and its assigned negative, all pairs. Columns
 three to six describe the absolute GC difference between the pair; L_1 is over the sixteen dinucleotide
 frequencies, on a 0 to 2 scale. The 0.05 column is a common yardstick and is the GC matcher's
 nominal tolerance; the dinucleotide matcher has no GC tolerance to hold.

arm	pairs	med. $ \Delta GC $	p90	p99	max	$ \Delta GC \leq 0.05$	med. L_1
GC-matched	456,734	0.0297	0.0495	0.1089	0.1486	94.8%	0.500
dinucleotide-matched	457,998	0.0198	0.0693	0.1485	0.4159	85.0%	0.220
bias-aware	456,734	0.1387	0.3169	0.4555	0.6733	20.4%	0.700

The GC matcher holds its nominal tolerance for 94.8% of pairs and its observed maximum,
 0.1486, sits at the 0.15 fallback cap. "Dinucleotide-matched" likewise means a median L_1 of 0.220
 210 rather than zero: the matcher improves composition distance 2.27-fold relative to the GC arm but

does not eliminate it. The bias-aware arm is not composition-matched at all, with a median GC gap 4.7 times the GC arm's.

The dinucleotide arm matches GC better than the GC arm does at the median, 0.0198 against 0.0297, and worse in the tail, a maximum of 0.4159 against 0.1486 and 85.0% within 0.05 against 94.8%. Both follow from the two matchers' objectives and neither is an error. GC content is very nearly a linear functional of the sixteen dinucleotide counts, and a nearest-neighbour search minimising L_1 over all sixteen pins GC incidentally and tightly for a typical pair. The GC matcher instead performs a tolerance test and accepts the first candidate inside it, which leaves the achieved gap spread over the acceptance interval rather than driven toward zero. The tails invert for the complementary reason. The GC matcher imposes a hard fallback cap and so cannot exceed 0.15, whereas the dinucleotide matcher imposes no maximum distance and assigns greedily: a positive reached late in its group, once the nearby candidates are used, takes the nearest remaining one however far off it is. The two arms therefore differ in which moment of the GC distribution they control, and the asymmetry the design rests on is in the degrees of freedom each protocol constrains, not in which achieves the smaller GC gap.

The pair counts differ by 1,264 between the dinucleotide arm and the other two. A positive is retained only when its matcher finds an acceptable negative, the two matchers fail on different windows, and the bias-aware arm inherits the GC arm's retained set; the resulting overlap between the composition-matched arms is quantified in the Limitations.

2.5 Cross-validation

Evaluation is grouped 5-fold cross-validation over whole chromosomes. The chromosome-to-fold map is frozen once for all datasets in both cell lines; the fifteen proteins measured in both lines are evaluated on the same chromosomes and a between-line difference cannot be a partition artefact. The map was chosen by random-restart hill climbing on peak counts only, never on labels or model output, minimising the summed squared deviation from equal mass per fold with every dataset weighted equally; it was required to beat round-robin assignment, to hold the median per-dataset maximum deviation at or below 0.05, and to place no more than half of any dataset's data in one fold. Fold i is the test fold, fold $i + 1 \bmod 5$ is validation and the remaining three train, so each fold is test once, validation once and training three times, and no additional data is spent on model selection.

Every window is scored exactly once out of fold and all AUROCs are computed on the pooled out-of-fold score vector. Pooling, instead of averaging per-fold AUROCs, makes the estimate invariant to fold size; the least balanced dataset places 0.42 of its data in a single fold.

Leakage between folds was audited by exact 32-mer sharing, not by clustered sequence identity. A 101 nt window contains 70 distinct 32-mers, and two unrelated sequences share one with negligible probability, and a shared 32-mer indicates common origin. Over all 94 datasets, holding fold 0 out, a median of 2.0% of held-out windows shared any 32-mer with a training window (maximum 24.3%), a median of 0.08% shared more than half of their 32-mers, and the median dataset contained

no window sharing more than 90%. Chromosome grouping therefore leaves a small residue of
 250 homologous sequence across folds, which is the quantity a sequence-identity clustering step would
 remove.

2.6 The composition baseline and the nested contribution

The composition baseline is a 19-column design matrix: three mononucleotide frequencies, fifteen
 dinucleotide frequencies and the Shannon entropy of the window’s base composition. One level is
 255 dropped from each frequency family because frequencies sum to one and retaining all of them makes
 the design singular; dropping a level does not change the space the block spans. Every column
 is standardised over the whole dataset, not within fold; the transform is label-free and identical
 in both terms of every comparison, and cannot manufacture a contribution. GC content is not
 a feature: it is the spanned combination C+G, and mononucleotide counts are almost, but not
 260 exactly, marginals of dinucleotide counts: over a window of length L the dinucleotide counts sum to
 $L - 1$ and recover each base’s count only up to whether the terminal base is that base. The linear
 block therefore has rank 18 and not 15, with a median 0.14% of each mononucleotide column’s
 variance lying outside the span of the dinucleotide columns (maximum 0.25% over 25 datasets), and
 the entropy column adds a nineteenth, nonlinear direction. What the two protocols constrain is
 265 nonetheless asymmetric by construction, and that asymmetry is what the design argument rests on:
 GC matching fixes a single linear functional of local composition, whereas dinucleotide matching
 fixes all fifteen degrees of freedom of the dinucleotide simplex, and leaves the baseline strictly less
 compositional variation to exploit. The baseline is refit on each protocol’s own windows and is never
 carried between protocols.

270 The nested contribution of a model is

$$\text{AUROC}(\text{composition} + \text{standardised model score}) - \text{AUROC}(\text{composition alone}).$$

Both terms were fitted by L_2 -penalised logistic regression ($C = 1$, lbfgs, at most 3000 iterations),
 once per fold on that fold’s training rows, and evaluated as pooled out-of-fold linear predictors on
 identical rows and folds. The model-score column is the out-of-fold score for every row, training
 rows included, so the nested model is never fitted on an in-sample score; had it been, the fitted and
 275 evaluated covariate would sit on different scales and the mismatch would grow with how much the
 underlying model overfits, which differs sharply between a 256-column penalised logistic and a fully
 fine-tuned transformer.

One channel remains open and is worth naming, because it is not removed by scoring out of
 fold. For test fold i the nested model is fitted on the rows of the other folds, and those rows carry
 280 scores from base models whose own training sets included fold i . Information about fold i therefore
 reaches the nested fit through the covariate, and the composition columns have no analogous route.
 The bias is one-directional, since it can only help the score column, and it should grow with how
 much the base model overfits. We do not remove it; we measure it. A model whose information the

baseline already contains has a true increment of zero, and what the estimator returns for such a
 285 model is reported in Section 3.9: it is this channel, together with the conditioning effect described
 there, that the floor quantifies.

The two were compared by DeLong’s paired estimator [12], in the $O(n \log n)$ midrank form of
 Sun and Xu [13]. The paired estimator is necessary here because the two score vectors are fitted
 on overlapping training data and evaluated on identical rows, so they are strongly correlated and
 290 treating them as independent would overstate the variance of their difference: the median DeLong
 standard error of the difference is 0.26 times what independence gives. Confidence intervals on a
 single dataset’s contribution are Wald intervals on the DeLong standard error.

Demler et al. [14] show that DeLong’s test is not valid for strictly nested models, because under
 the null the added predictor’s contribution degenerates. Two things bound the exposure here. The
 295 panel-level intervals, which carry every headline, do not use the DeLong standard error at all; they
 are protein-clustered bootstrap intervals over per-dataset point estimates. And where the DeLong
 standard error is used, for the per-dataset significance counts, it is conservative relative to the
 recommended alternative: a penalised likelihood-ratio test on the same fits calls 89 of 94 datasets
 significant in the GC arm against DeLong’s 80. A third feature of the design might have cut the
 300 other way and does not: negatives are matched 1:1 to positives, so the rows are paired, while DeLong
 treats the two classes as independent samples. Resampling matched pairs instead reproduces the
 DeLong standard error to within 1% (median ratio 0.99 over 30 datasets), so the matching needs no
 correction.

Both arms are evaluated out of fold. An in-sample composition AUROC compared against an
 305 out-of-fold model AUROC flatters composition sufficiently to reverse the comparison on individual
 datasets.

2.7 Models

4-mer logistic regression. Raw 4-mer counts over the RNA sequence (98 per window, 256
 columns) into L_2 -penalised logistic regression ($C = 1$). One model per fold, out-of-fold score taken
 310 as the linear predictor. The k -mer order is 4, and the choice is not load-bearing: rebuilding the
 primary contrast from raw sequence at each order gives +0.0311 at $k = 3$, +0.0397 at $k = 4$, +0.0397
 at $k = 5$ and +0.0355 at $k = 6$, every interval excluding zero, with the dinucleotide arm larger in 90,
 88, 89 and 91 of 94 datasets respectively and 82 of 94 datasets positive at every order. The $k = 5$
 minus $k = 4$ difference is +0.0001 (paired Wilcoxon $p = 0.84$). Order 4 is the smallest that spans
 315 the dinucleotide baseline strictly, and the contrast is smallest at $k = 3$, so the reported figure is not
 the most favourable order available.

Convolutional network. A DeepBind-style architecture [19]: convolution (4 to 16 channels,
 width 12), ReLU, max-pool 4, convolution (16 to 32, width 8), ReLU, global max-pool over position,
 dense 32 to 64, ReLU, dropout 0.5, dense 64 to 1. 7089 parameters, all trained from scratch. Global
 320 max-pooling makes the model position-invariant, which is necessary here because the discriminative

signal is not at the window centre. The mutual information in bits between the base at each of the 101 positions and the label, over every window and with no model fitted, puts the most informative position a median of 23 nt from the midpoint, interquartile range 11 to 29 nt and maximum 48. It lies within 5 nt of the midpoint in only 13 of 94 datasets. The midpoint itself carries 0.0058 bits
325 against 0.0136 at the peak, so less than half. A position-sensitive architecture would therefore be fitting a location that moves from protein to protein.

SpliceBERT. The pretrained nucleotide language model of Chen et al. [21], `multimolecule/splicebert`, fully fine-tuned; 19.78 M parameters, all trainable. The classification head is a mean pool over real nucleotide tokens, masking classification, separator and padding tokens, then dense
330 to 128, ReLU, dropout 0.3, dense to 1. Weights are baked into the container image at build time.

Both neural models: AdamW, weight decay 0.01, batch size 32, binary cross-entropy, single precision, at most 12 epochs with early stopping on validation AUROC at patience 4, best checkpoint restored before test scoring. Learning rate 10^{-3} for the head and 3×10^{-5} for a fine-tuned encoder. Nominal epochs overstate the training performed: the median best epoch for SpliceBERT was
335 3 and most runs stopped by epoch 6 or 7, whereas the convolutional network typically ran the full 12. Initial weights were drawn before the seed was set, so initialisation is unseeded across all 2830 committed fold-runs. The measured cost is a per-dataset standard deviation of 0.006 to 0.010 in the nested contribution, inducing about 0.001 on a 94-dataset mean against a reported interval half-width of about 0.008. The convolutional and transformer arms were trained on different
340 accelerators, which changes floating-point summation order and is covered by the same measurement.

2.8 Inference

Because fifteen of the 79 proteins contribute two datasets each and the within-protein correlation of the primary contrast is 0.92, every headline interval is a percentile interval from a bootstrap that resamples *proteins* and takes all datasets of each sampled protein, 4000 draws. Dataset-level
345 intervals are 1.03 to 1.23 times narrower and are not reported as headline values.

Two analyses are confirmatory and are the only ones we adjust. The first is the primary contrast, the difference in nested contribution between protocols, which the study was built to estimate. The second is the recommendation test of Section 3.11, whose two falsifying criteria were fixed before the external data were scored and whose interval is Bonferroni-corrected over the three protocol pairs.
350 Everything else here is exploratory. That includes the baseline attribution, the family partition, the order-three contrast, the within-panel reversal and the caliper-matched nulls, all of which we report as point estimates with intervals rather than as decisions, and none of which is adjusted for the number of analyses performed. Readers should treat an exploratory interval that just excludes zero accordingly.

Per-dataset significance counts are also reported at a design effect applied to the DeLong standard error, which is intended to cover two things the estimator does not: that it conditions on two score vectors as fixed when both are fitted, and that it treats spatially clustered windows as independent.
355

We measured both components instead of assuming them (`scripts/design_effect.py`). Resampling whole genomic blocks inflates the standard error of the contribution by a median factor of 1.100, and the factor is insensitive to block length, moving only to 1.104 at 100 kb and 1.138 at 1 Mb, so the objection that a 10 kb block is smaller than a gene does not bite. Permuting the model score within fold and refitting gives 1.048. The measured product is therefore 1.15. We report the counts at that measured value as primary, on the same principle applied everywhere else here, that a measurement supersedes a figure chosen before it was taken. The value fixed in advance was 1.35, and it is retained as a conservative sensitivity: 75 of 94 datasets are significant at 1.15 and 72 of 94 at 1.35, against 80 of 94 with no inflation at all. The choice between the two moves three datasets and no conclusion. This is one of two quantities in the paper that the verification harness cannot check from the released evidence, because the block bootstrap needs the genomic coordinates that the published per-window score files do not carry.

2.9 Reproducibility

All published values are reproduced offline by a single command against committed evidence. The harness runs 937 numeric assertions against a frozen expectations file, and the harness asserts that no fewer than that number ran, so a check cannot silently skip. Two assertions are stronger than regression gates: 285 published AUROCs are rebuilt from committed per-example scores to a maximum absolute difference of 2.2×10^{-16} , and the headline contrast is rebuilt from raw sequence to 1.2×10^{-6} . Per-window out-of-fold scores for all three model classes are committed, so every cell of the model-class comparison is recomputable from the repository alone, and is not asserted against a hand-entered table.

Three devices guard against platform-dependent results: dinucleotide distances are computed on integer counts, candidate ties are broken by genomic position, and the panel sort is stable. Each was added after an observed discrepancy between CPU architectures or between runs.

Analyses used NumPy [22], SciPy [23], scikit-learn [24], PyTorch [25] and Matplotlib [26].

3 Results

3.1 The negative-set protocol changes measured contribution more than five-fold

Across 94 paired datasets the same 4-mer model gave three different accounts of its own contribution depending only on how the negative windows were built (Table 2, Figure 1).

Replacing GC-matched with dinucleotide-matched negatives reduced the 4-mer’s own AUROC in all 94 datasets (paired Wilcoxon $p < 10^{-15}$), from 0.7981 to 0.6879, while raising the nested contribution from +0.0265 to +0.0663. The contrast in contribution was +0.0398 (95% CI +0.0325 to +0.0477, protein-clustered; positive in 88 of 94 datasets). Evaluated by its own AUROC the model appears to have deteriorated; evaluated by contribution it appears to have improved by a factor of 2.50 in panel means, or 2.94 as the geometric mean of the 88 per-dataset ratios in which

Table 2: Composition-only AUROC, the 4-mer’s own AUROC, the AUROC of the two together, and the nested contribution, under three negative-set protocols. The nested contribution is the composition-plus-4-mer column minus the composition-alone column by construction. The 4-mer’s own AUROC is what a study would report as its headline. Model, folds and estimator are identical across arms; positives are identical in 10 of 94 datasets and share a median Jaccard of 0.997, as the Limitations set out. $n = 94$ datasets.

protocol	composition	4-mer alone	composition + 4-mer	contribution
dinucleotide-matched	0.6274	0.6879	0.6937	+0.0663
GC-matched	0.7827	0.7981	0.8092	+0.0265
bias-aware	0.8248	0.8169	0.8370	+0.0122

both arms are positive. Across the three arms the contribution spanned 5.42-fold (95% CI 4.43 to 6.58, protein-clustered).

395 The bias-aware protocol did not behave as we predicted. Because both classes are genuine crosslink sites, we expected composition alone to fall towards 0.5 and the contribution to rise. Neither occurred: composition alone was the highest of the three arms at 0.8248, the AUROC of composition plus the model was the highest at 0.8370, and the contribution was the lowest at +0.0122, or 0.53 times the GC arm (geometric mean over the 83 datasets positive in both arms; the
400 ratio of panel means is 0.46). Part of this is transcript-region mix and not site composition; the next subsection separates the two. The absolute value carries a further qualification established in Section 3.9: this arm’s +0.0122 is close to what the estimator returns for a model that contributes nothing, so it is the ordering and not the level that the comparison rests on.

The two AUROC rankings of the three protocols agree with each other, and the ranking by
405 contribution is their exact reverse. Harder discriminations do reveal more of what a model adds; what does not follow is that a bias-aware protocol is the harder one. The starkest form of this is that the 19-feature composition baseline beat the 4-mer outright in 14 of 94 datasets under dinucleotide matching, 29 of 94 under GC matching, and 53 of 94 under the bias-aware protocol, where it also beat the model on the panel mean. Under GC matching the contribution was significantly positive
410 in 80 of 94 datasets on unadjusted marginal intervals, 79 of 94 after Benjamini-Hochberg control within the panel, and 75 of 94 once the measured design effect of 1.15 is applied, or 72 of 94 at the more conservative 1.35 fixed in advance (Methods).

The direction of the GC-versus-dinucleotide contrast is implied by the design. The composition baseline reads local composition, and the two protocols constrain it asymmetrically: GC matching
415 fixes a single linear functional of that composition, whereas dinucleotide matching fixes all fifteen degrees of freedom of the dinucleotide simplex. The arm that constrains more must leave the baseline weaker. Only the magnitude of that contrast is informative. The position of the bias-aware arm is not implied by the same argument. That protocol constrains none of those degrees of freedom, and its negatives are not composition-matched in any respect (median absolute GC difference 0.1387,
420 against 0.0297 for the GC arm). It nonetheless has the highest composition baseline of the three.

We bounded two artefacts against matched negative controls, drawn to match region and GC,

with twenty control seeds per dataset per arm. Both tests are on the GC-versus-dinucleotide contrast and on subpanels, and each percentage below is taken against its own subpanel’s contrast and not against the +0.0398 headline. Strand misassignment, which arises because a matched negative inherits its positive’s strand label, accounted for -0.0055 (CI -0.0089 to -0.0022) of the +0.0378 contrast on the 40 datasets where the test is possible, leaving 85.4%. Negatives drawn from untranscribed loci accounted for -0.0043 (CI -0.0069 to -0.0018) of the +0.0414 contrast on the 84 datasets with expression data, leaving 89.7%. The two mechanisms are correlated, and 0.854 and 0.897 should not be multiplied. On the transcription control, 40.1% of negatives lie in untranscribed loci and 67.6% are not transcribed on the strand the window is labelled with, but both fractions are balanced between the two arms, the first to within 1.2×10^{-5} ($p = 0.64$), and a confound present equally in both arms cannot produce a difference between them. Counting all negatives, 42.8% of GC-matched negatives carry the strand of their own gene against 47.4% in the dinucleotide arm, which is the higher of the two in 37 of 40 datasets. On the paired per-dataset fractions a two-sided Wilcoxon signed-rank test gives $p = 5.0 \times 10^{-9}$; the count and the test are separate statements, and the p belongs to the second. So the spurious directional cue is stronger in the arm with the smaller contribution and acts against the reported direction.

The strand test admits a stronger reading, which we report here so that it is not left to be found. Within the panel, regressing the contrast on the sense-strand fraction gives a slope of -0.187 , and extrapolating to a fully sense-correct arm gives -0.046 , a sign reversal. That extrapolation carries no weight: over the negatives whose strand is determinable, the fraction spans only 0.433 to 0.615 across datasets, so the leverage available is 0.18 and the extrapolation runs far outside it, the interval on the extrapolated value is -0.189 to $+0.072$, and the direct test, the correlation between the sense fraction and the contrast, is -0.240 ($p = 0.136$) and null. The point estimate is nonetheless the one number in this paper whose naive reading reverses the headline. None of these three controls covers the three-arm ordering, only the composition-matched pair; Section 3.2 is the corresponding bound for the bias-aware arm.

3.2 Part of the bias-aware arm’s baseline is transcript region, and the ordering survives removing it

The bias-aware arm’s position is the one result here that the design does not imply, and it is the one most exposed to a design asymmetry, and there is one. Both composition-matched protocols draw candidates of the same region class as the positive; region carries nothing: a classifier given only the five region dummies scores exactly 0.5000 in both arms, in all 94 datasets. The bias-aware protocol matches fold only, and there region alone separates the two classes at a median AUROC of 0.748, above 0.70 in 67 of 94 datasets, with the positive and negative region marginals differing at a median total variation distance of 0.419 (Table 3). The effect is graded, not uniform: the more region separates in a dataset, the more its bias-aware baseline exceeds its GC baseline (Spearman $+0.287$, $p = 0.005$) and the larger its contribution deficit against that arm (Spearman -0.253 , $p = 0.014$).

Both statements rest on region labels, and the label a matched negative carries is the region

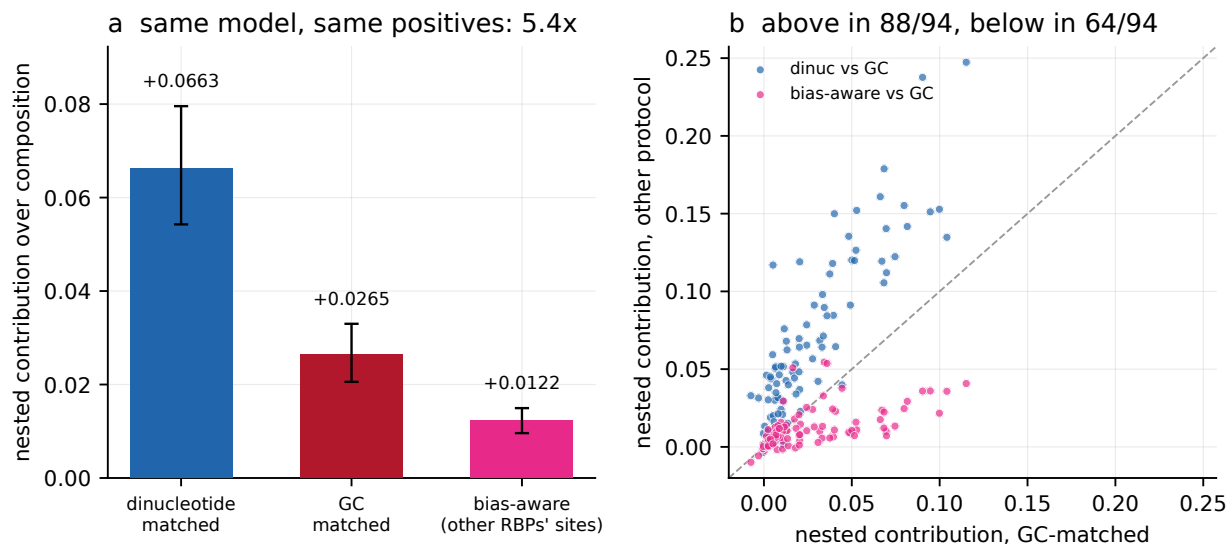


Figure 1: Nested contribution of a 4-mer model under three negative-set protocols, $n = 94$ datasets. (a) Panel means by protocol, +0.0663, +0.0265 and +0.0122, a 5.4-fold span for one model on one set of positives; bars are 95% percentile intervals from a bootstrap over the 79 proteins, taking all datasets of each sampled protein (4000 draws), as everywhere else in this paper. (b) Per dataset, the contribution under each of the other two protocols against its value under GC matching. Dinucleotide matching lies above the diagonal in 88 of 94 datasets and the bias-aware protocol below it in 64 of 94. The same contributions plotted against the composition-only AUROC each protocol leaves are Figure 3a, where that gradient is the subject.

pool the matcher drew it from rather than a classification of the window drawn. Merged per-region intervals overlap, because a position can be a coding exon of one isoform and a 3' UTR of another, and the two need not agree. Recomputing the annotation rule from the same GENCODE index reproduces the committed label for every one of the 2,251,282 positives across all three arms and differs for 10.33% and 10.17% of the GC and dinucleotide arms' negatives. It differs for 0.00% of the bias-aware arm's, whose negatives are other proteins' positives and so carry genuine classifications.

That distinction matters for the exact 0.5000 above, and both readings are reported. For the region the matcher enforced it is exact, and that is the right quantity for what the matcher did. If instead both classes are re-annotated by one rule applied afresh, the composition-matched arms sit at a median 0.5483 and 0.5452 rather than at a half: close to uninformative on region but not exactly so. The bias-aware arm's 0.7484 is unchanged by that re-annotation, since its labels were already classifications.

The asymmetry itself does not depend on the annotation rule, and we checked this against three alternatives a careful analyst could have chosen instead: the priority order reversed, coding classes first, and majority overlap, which ignores the midpoint and takes the region covering most of the window. These change a fraction 0.4122, 0.1080 and 0.1119 of all labels. The gap between the bias-aware arm and the composition-matched arms is positive under all five annotations, ranging from +0.2484 on the enforced labels to +0.1055 under the reversed priority, which is the least

favourable and also the least defensible of the four rules, since it prefers the intron call over the more specific exonic one. The bias-aware arm’s own figure is not rule-invariant either, running
480 from 0.6622 to 0.7617. So region separates the bias-aware classes and does close to nothing in the matched arms under every rule tried, while the magnitude of that difference is annotation-dependent by roughly a factor of two.

We therefore rebuilt the arm with region matched, stratifying the donor draw on region class instead of drawing uniformly within the fold. Every other element of the construction is unchanged,
485 and the rebuild keeps all 456,734 pairs, with no pair lost to subsampling. It achieves the match exactly, at a region-only AUROC of 0.5000 and a total variation distance of 0.0000. The composition baseline falls from 0.8248 to 0.8052 and the contribution from +0.0122 to +0.0092, so 46% of the arm’s baseline excess over the GC arm is region mix and the rest is not. The ordering is unchanged: the region-matched arm still carries the highest composition baseline of the three and still yields
490 the lowest contribution.

That repair was available for the 4-mer alone until now, which left the obvious objection open: a convolutional network or a pretrained language model might depend on transcript region in a way a bag of 4-mer counts does not, and the arm whose position the design does not imply was corrected for only one model class of three. We therefore trained both neural models on the region-matched
495 arm across all 94 datasets and five folds, on the same code path and hardware as the published sweeps, 940 fold-runs in total. The 4-mer column recomputed here reproduces the independently computed value above to 1.1×10^{-16} , and the neural columns beside it are measured on the same arm.

The answer does not change for any model class, and the correction moves the result the wrong
500 way for a confound. With region matched the contributions are +0.0092, +0.0100 and +0.0352 for the 4-mer, the convolutional network and SpliceBERT; the first two lie below the estimator floor of Section 3.9 and should be read as upper bounds rather than as measured increments, and the region-matched arm still gives the smallest of the three protocols for every one of them. The span against the dinucleotide arm *widens* in all three cases, from 5.42 to 7.20, from 7.42 to 8.40 and
505 from 3.72 to 4.93. A result driven by the region confound would have narrowed when the confound was removed. SpliceBERT loses the most to the repair, −0.0114 against −0.0030 and −0.0013: the pretrained model was the one leaning hardest on transcript annotation, which is the direction the objection anticipated even though the conclusion is unaffected.

One ordering does move, and it confirms rather than contradicts what we report above. In the
510 published bias-aware arm the 4-mer leads the convolutional network by +0.0010 with an interval spanning zero, which we describe as the protocol destroying the ranking between them rather than reversing it. Matching region flips the sign of that difference, to +0.0100 against +0.0092. A ranking that changes direction under a correction this small is a ranking the arm does not determine.

Both alternative constructions agree. Reweighting the donors already drawn, instead of redrawing
515 them, gives 0.8017 and +0.0062 from a construction that discards 30% of the rows. And on the half of the panel where region separates least, which requires no rebuild at all, the bias-aware baseline is

still the higher (0.8184 against 0.7962 for the same datasets’ GC arm) and the contribution ratio is still below one at 0.641.

We do not propose this arm as a fourth protocol. Horlacher et al. [4]’s negative-2 does not match region either, and a region-matched version of it is not the design the field proposed. Its purpose is to measure how much of the arm’s baseline is transcript annotation and how much is binding-site composition.

Table 3: What the region label alone achieves in each arm, and the bias-aware arm rebuilt with region matched. Region is matched by construction in both composition-matched arms and free in the bias-aware arm. The rebuilt arm is a diagnostic, not a fourth protocol; the reweighted row is a second estimate from a different construction, which discards 30% of the rows. $n = 94$ datasets.

arm	pairs kept	region AUROC	region TV	composition	contribution
dinucleotide-matched	100%	0.5000	0.0000	0.6274	+0.0663
GC-matched	100%	0.5000	0.0000	0.7827	+0.0265
bias-aware	100%	0.7484	0.4192	0.8248	+0.0122
bias-aware, region-matched	100%	0.5000	0.0000	0.8052	+0.0092
the same, by reweighting	70%	0.5026	0.0043	0.8017	+0.0062

The explanation for the arm’s position is therefore narrower than that distinct RBPs occupy compositionally distinct sites. Distinct RBPs occupy distinct transcript regions, which the composition baseline reads through the base composition characteristic of each region, and they also occupy compositionally distinct sites within a region. Both contribute, and only the second is about binding-site composition.

One alternative explanation for the arm’s deficit is not compositional at all and has to be excluded rather than argued away. Many RBPs co-bind the same transcripts, and a negative drawn from another protein’s binding site may be a site the target also binds, which is a mislabelled positive rather than a hard negative, and label noise of that kind depresses any model’s measured contribution on its own. The construction excludes a donor window within 500 nt of the target’s own positive windows, which is not the same as excluding it from the target’s peaks, because a positive is one 101 nt window on a peak midpoint and a wide peak extends beyond it. We measured the residual directly: the fraction of bias-aware negatives whose midpoint falls inside a called peak of the target protein has median 0.0000, mean 0.0001 and maximum 0.0029 across the 94 datasets. At three negatives in a thousand at worst, there is not enough mislabelling present to account for the arm’s deficit of 0.0143 against the GC arm.

The direction is wrong for it as well. Sorting datasets by that residual, the third with the most overlap has the *smallest* deficit, -0.0078 against -0.0190 for the third with the least, a difference of $+0.0112$ in the opposite direction to what co-binding predicts. The across-dataset correlation is $+0.228$ ($p = 0.027$), also the wrong sign, though with a residual this near zero throughout we would not read much into its size either. The claim we make is the negative one: co-binding is too rare here to be the explanation, and what variation it has does not point the way the explanation requires.

3.3 No standard reparameterisation removes the protocol dependence

AUROC is compressive near unity: a contribution measured against a baseline of 0.82 is arithmetically smaller than the same contribution measured against 0.63, and part of the 5.42-fold span comes from the scale itself (Figure 2).

We compared eight rescalings of the same 282 protocol-dataset cells, and they are of two kinds. Six take the increment on a transformed AUROC scale: the raw difference, Somers’ D [16], the binormal d' , and the logit, arcsine and complementary log-log links. Two are not transformations of the contribution at all but normalisations of it by a baseline-dependent quantity, namely division by the headroom $1 - c$ and by the excess over chance $c - 0.5$. Both are kept in the comparison because they are what a reader reaches for. Five of the eight reduced the span without closing it, two are controls that return the raw span by construction, and one increased it. Somers’ D is an affine map of AUROC and so returns 5.42 exactly, alongside the raw difference, which is the reference. Links that stretch the upper tail reduced the span in proportion to how far they stretch it: arcsine 3.99, complementary log-log 3.94, binormal d' 3.18, logit 2.72. The smallest span, 2.00 (CI 1.67 to 2.46), was obtained by dividing the contribution by the headroom of its own baseline, 1 minus the composition AUROC. Normalising by the excess over chance instead increased the span to 18.2, because that denominator varies with protocol and approaches zero on some datasets.

A span of 2.00 should not be compared against unity. The ratio of the largest to the smallest of three sample means is bounded below by one and is upward-biased, so some span is expected under any sampling. Simulating equal true arm means while preserving the observed between-arm dataset pairing gave a null distribution with median 1.064 and 95th percentile 1.175, so the residual span lies outside what sampling alone produces.

Division by headroom belongs to a family of baseline-dependent normalisations, the $p = 1$ member of $g/(1 - c)^p$, where g is the contribution and c the composition AUROC, and the minimum of the sweep above falls on the $p = 1$ member. Over the family the span falls to 1.004 at $p = 1.544$ (Table 4, first row), which reads as an exponent that equalises the three protocols. We retract that reading. It does not survive the choice of aggregation, and the reason is the denominator.

Every span in this paper aggregates by taking, for each arm, the mean over datasets of that dataset’s ratio, and then the largest of the three arm means over the smallest. When $p = 0$ the ratio is just the contribution and the aggregation is innocuous. As p grows it is not, because $1 - c$ is small on many cells: it reaches 0.0267 in the bias-aware arm and 0.0335 in the GC arm, 62 of the 282 cells fall below 0.15 and 10 below 0.05, so a mean of $g/(1 - c)^p$ is dominated by a handful of high-baseline datasets, and increasingly so with p . Aggregating in any way that is not a mean of a ratio removes the effect. The median of the per-dataset ratios bottoms out at 1.238 at $p = 2.17$, and the ratio of the panel means, $\bar{g}$ over $\overline{(1 - c)^p}$, at 1.278 at $p = 2.64$. Neither approaches unity anywhere in the family. The equalising exponent is therefore a property of one aggregation interacting with a near-zero denominator, not a coordinate in which these three protocols agree.

Even taken at face value the exponent describes this benchmark only. Fitted to the two negative sets of Horlacher et al. [4] the equalising exponent is 3.649, and our exponent leaves their benchmark

Table 4: Fold span across the three protocols for $g/(1 - c)^p$, where g is the nested contribution and c the composition-only AUROC, under three ways of aggregating the same 282 cells. Only the mean of per-dataset ratios, the aggregation used throughout this paper, approaches unity; it is also the one a denominator of 0.027 can dominate. The final column is the minimum over p on $[0, 6]$ and the exponent attaining it.

p	0	0.5	1.0	1.25	1.544	1.75	2.0	min	at p
mean of ratios	5.42	3.43	2.00	1.48	1.004	1.34	1.95	1.004	1.54
median of ratios	5.09	3.92	2.61	2.34	1.70	1.37	1.29	1.238	2.17
ratio of means	5.42	3.62	2.55	2.17	1.82	1.61	1.44	1.278	2.64

585 at 2.340 against the 2.381 it started from. A normalisation strong enough to equalise one benchmark therefore has to be refitted on the next.

Extending the sweep to the other two model classes qualifies the headline figure. The floor over the eight reparameterisations is 2.00 for the 4-mer, 2.78 for the convolutional network and 1.65 for SpliceBERT: no model class reaches protocol independence among fixed coordinates and all three exceed the equal-means null, but 2.00 is the 4-mer’s floor, and each model class has its own.
590 The fitted exponent also performs worse for the neural models, reaching 1.241 for the convolutional network and 1.133 for SpliceBERT against 1.004 for the 4-mer, and the equalising exponent itself differs by a factor of 1.21 between model classes measured on the same benchmark (1.544, 1.793 and 1.485).

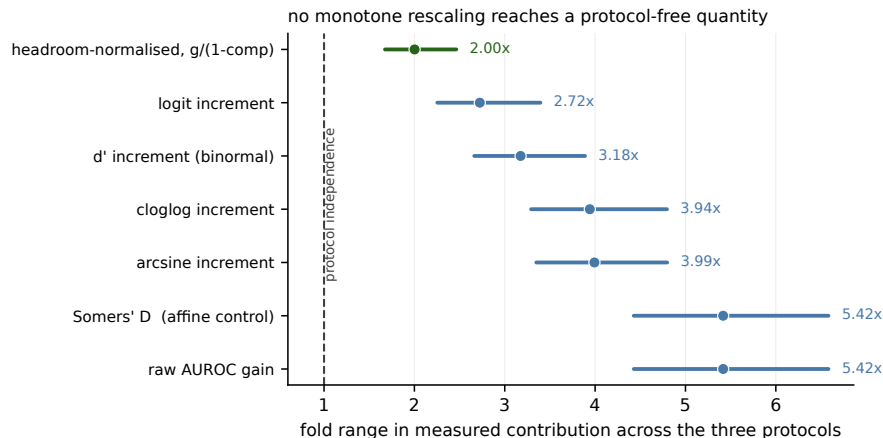


Figure 2: Fold span in nested contribution across the three protocols under eight monotone rescalings of the same 282 protocol-dataset cells. Seven of the eight are shown; normalising by the excess over chance is omitted because its span of 18.2 is off scale. Points are the span and bars are 95% percentile intervals from a bootstrap over the 79 proteins (4000 draws); the dashed line marks protocol independence. Somers’ D is an affine map of AUROC and returns the raw value by construction, serving as an internal control. The smallest span, 2.00 (1.67–2.46), is obtained by dividing by the headroom of the baseline.

3.4 The composition baseline explains more of the dependence than the protocol label, for a k -mer model

How difficult a protocol looks and how difficult it is for composition alone are close to the same thing on this panel. Pooled over the 282 protocol-dataset cells, the 4-mer’s own AUROC and the composition-only AUROC correlate at Pearson +0.894 (protein-clustered interval +0.854 to +0.926; Spearman +0.892), and within arms at +0.911, +0.624 and +0.956 for the GC, dinucleotide and bias-aware protocols. A protocol that is hard for a 4-mer is hard for composition, which is why the composition baseline is a candidate for the thing through which the protocol acts.

Pooled over the same 282 cells the contribution was inversely related to the composition baseline (Spearman -0.600 , protein-clustered interval -0.698 to -0.495), so much of what a protocol does is set how much room the baseline leaves (Figure 3). We quote an interval and not the nominal p , which treats the 282 cells as independent when each dataset appears once per arm and fifteen proteins contribute two datasets, so between three and six correlated cells share a protein.

Regressing contribution on a quadratic in the composition baseline over all 282 cells and then adding protocol indicators, the protocol label contributed 1.0% of variance given the baseline and the baseline contributed 11.0% given the label. Neither increment is corrupted by collinearity, since the baseline recovers the arm with only 58.5% accuracy against a chance rate of 33.3% and the protocol dummies regressed on the baseline curve give variance inflation factors of 1.11 and 1.32. The two increments answer different questions, though, and their ratio is not a decomposition of one variance into two competing causes: the first is a protocol-specific offset after a smooth baseline curve, the second a pooled within-arm gradient. The protocol’s actual manipulation is within dataset, where the dinucleotide baseline is below the GC baseline in 94 of 94 datasets, so the first increment is estimated almost entirely from between-dataset variation. With dataset fixed effects the two increments become 0.33% and 6.33%.

The bias-aware protocol usually raises the baseline relative to GC matching, but lowers it in 27 of 94 datasets, which permits a within-panel test. Where it raised the baseline its contribution deficit was -0.0212 (CI -0.0269 to -0.0157); where it lowered the baseline the deficit was abolished and the point estimate changed sign, at $+0.0028$ (CI -0.0000 to $+0.0063$, a dataset bootstrap whose interval therefore just includes zero; the protein-clustered interval is $+0.0002$ to $+0.0063$ and excludes it). Within datasets, the change in baseline and the change in contribution were related at Spearman -0.664 ($p = 3 \times 10^{-13}$).

Matching on the baseline directly is possible only across datasets, because the dinucleotide baseline is below the GC baseline in all 94 datasets and the two are therefore perfectly rank-confounded within a dataset. Under nearest-baseline matching, with a caliper of 0.02 AUROC, the $+0.0398$ contrast fell to -0.0087 (CI -0.0265 to $+0.0122$, $n = 41$ pairs; the interval is -0.0284 to $+0.0219$ when the bootstrap is clustered on the reused GC partners). This is not a clean test of the contrast, for two reasons that point the same way. The retained pairs are drawn from opposing tails of their respective distributions: mean baseline 0.6875 against a panel mean of 0.6274 for the dinucleotide members, and 0.6904 against 0.7827 for their GC partners. The 41 comparisons are

served by 24 distinct GC partners, one of them reused eleven times, and proteins are not matched; the common support for this pair is 0.073 AUROC wide over 33 of 188 cells. More fundamentally, the composition baseline is the mediator this paper proposes for the protocol’s effect, so matching on it estimates a controlled direct effect, and a null direct effect is what the mechanism predicts, not evidence against it.

The GC-versus-bias-aware pair overlaps by 0.212 AUROC over 130 of 188 cells, where the same design is better conditioned. There, with a caliper of 0.01, the bias-aware arm retained a deficit of -0.0081 (CI -0.0130 to -0.0036) under nearest-baseline matching and -0.0043 (CI -0.0077 to -0.0012) as a within-dataset intercept at zero baseline shift. Decomposed by pair, the protocol label contributed 0.05% of incremental variance for GC versus dinucleotide and 3.40% for GC versus bias-aware, so the pooled 1.0% averages a comparison that is not identified with one that is.

Within arms, the relation between baseline and contribution was present for composition-matched negatives and not detectable otherwise, on protein-clustered intervals: Spearman -0.545 (-0.704 to -0.371) in the GC arm, -0.460 (-0.615 to -0.280) in the dinucleotide arm and -0.122 (-0.331 to $+0.082$) in the bias-aware arm, where the interval includes zero. Read on that axis this would suggest two protocol families rather than three interchangeable protocols, with a residual of 0.004 to 0.008 AUROC between them. It does not survive a change of axis. The statistic correlates a difference with its own subtrahend, which is legitimate but not invariant to which term is placed on the horizontal axis, and one arm changes family under the change. Putting the composition-plus-model AUROC there instead gives -0.315 (GC), $+0.352$ (dinucleotide) and $+0.031$ (bias-aware); at the midpoint of the two, -0.444 , $+0.025$ and -0.046 . Under either alternative the dinucleotide arm behaves like the bias-aware arm and not like the GC arm, so the family partition that survives every choice is the GC arm against the two others, not the composition-matched pair against the bias-aware arm. The variance ratio explains why: $\text{var}(\text{contribution})$ over $\text{var}(\text{baseline})$ is 0.61 in the dinucleotide arm against 0.09 and 0.03 in the other two, so neither axis is uncontaminated there.

These analyses do not generalise across model classes (Table 5). For the convolutional network the ordering reverses, the protocol label contributing 4.60% of variance against the baseline’s 3.47%, a narrower gap than the 4-mer’s in the other direction. For SpliceBERT the within-arm relation is strong in all three arms, including the bias-aware arm at -0.568 where the 4-mer shows -0.122 , so no two-family structure is apparent in that measurement. The variance attribution, the within-panel reversal and the family structure hold for the 4-mer measurement and should not be read as properties of the quantity.

3.5 Shuffling removes the baseline entirely, and the constructions in use leave it free

Before adding a fourth arm it is worth asking what this literature actually does, because we assumed an answer and it was wrong. Across seven widely used methods and benchmarks (Table 6), five construct negatives by relocating genomic intervals: GraphProt shuffles the coordinates of bound sites within genes carrying at least one site, the RBP-24 dataset behind several later models does

Table 5: Attribution of variance in the nested contribution, and the within-arm relation between baseline and contribution, by model class ($n = 94$).

model	incremental R^2			within-arm Spearman		
	protocol	base	base protocol	dinuc	GC	bias-aware
4-mer	1.00%		11.05%	−0.462	−0.545	−0.122
CNN	4.60%		3.47%	−0.185	−0.410	+0.075
SpliceBERT	1.12%		25.62%	−0.604	−0.767	−0.568

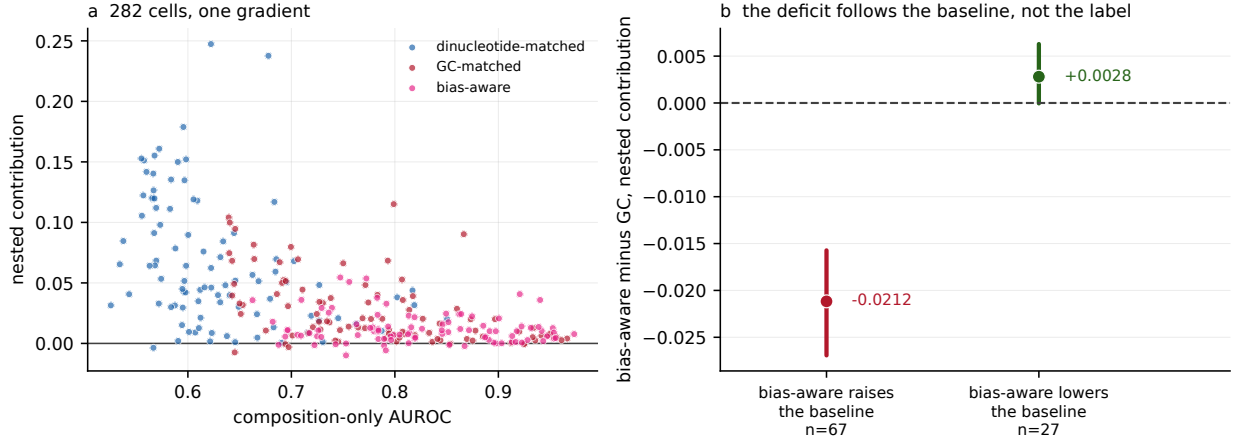


Figure 3: (a) All 282 protocol-dataset cells: nested contribution against the composition-only AUROC left by the protocol, by protocol (pooled Spearman -0.600 , $p = 6 \times 10^{-29}$). (b) Datasets partitioned by whether the bias-aware protocol raised or lowered the composition baseline relative to GC matching ($n = 67$ and 27); points are stratum means and bars are 95% percentile intervals from a bootstrap over datasets (4000 draws). The contribution deficit is abolished where the baseline falls.

the same with BedTools, DeepCLIP and RNAProt sample unbound regions of the same gene or transcript, and RBPsuite 2.0 shuffles to non-peak regions of the same transcript. Two draw negatives from other proteins’ binding sites. None of the seven permutes the positive’s own sequence as its primary construction; sequence shuffling appears only as an option for FASTA input in RNAProt and as an alternative in DeepCLIP. The dominant construction is therefore positional, and it leaves composition entirely unconstrained, which is why the composition baseline is free to vary across protocols in the first place.

None of the seven reports a composition-only AUROC beside its headline. DeepCLIP comes closest, observing a significant relationship between GC content and its own AUROC ($p = 7.89 \times 10^{-6}$) and attributing it to crosslinking bias rather than treating it as a baseline to report.

A sequence-permuting arm is still worth building, as the limiting case rather than as common practice. Dinucleotide-preserving shuffling is the construction Tourne et al. [5] indicts, and it is not a matching procedure at all. Because each negative is a permutation of its positive with dinucleotide counts held exactly, mononucleotide frequencies, dinucleotide frequencies, GC content and sequence

Table 6: Negative-set construction and baseline reporting in seven widely used methods and benchmarks. Coordinate: negatives are genomic intervals relocated within genes or transcripts that carry a positive. Other RBPs: negatives are binding sites of different proteins. The composition-baseline column records whether a composition-only AUROC is reported alongside the method’s headline AUROC.

source	year	negatives	composition baseline
GraphProt	2014	coordinate	no
RBP-24 dataset	2018	coordinate	no
pysster	2018	other RBPs	no
DeepCLIP	2020	coordinate	no
RNAProt	2021	coordinate	no
Horlacher et al. benchmark	2023	other RBPs	no
RBPsuite 2.0	2025	coordinate	no

entropy are identical between the two members of every pair. The nineteen-feature composition baseline takes the same value on both.

We built that fourth arm on the same 94 datasets, the same positives and the same folds (Table 7). The construction held: dinucleotide counts were preserved for every pair on the panel, and the composition feature vector was identical to nine decimal places in 100.00% of pairs. The composition baseline is therefore not approximately uninformative but exactly so, at 0.5000 in every dataset, since no threshold on a tied feature separates the classes. The 4-mer’s nested contribution over it is +0.2523, and it equals the model’s own standalone AUROC less a half to within 0.0006: with an uninformative baseline the increment is not an increment over anything, it is the model’s apparent performance relabelled.

Table 7: Four negative-set constructions, the 4-mer logistic regression, $n = 94$ datasets with identical positives and folds. The three matched arms are Table 8’s first row; the shuffled arm generates each negative from its own positive.

construction	composition alone	contribution	standalone AUROC
dinucleotide-shuffled	0.5000	+0.2523	0.7528
dinucleotide-matched	0.6274	+0.0663	
GC-matched	0.7827	+0.0265	
bias-aware	0.8248	+0.0122	

Two consequences follow, and they point in opposite directions for the paper’s own claims.

The first widens the headline. Across the three matched protocols the contribution spans 5.42-fold; adding the shuffled arm makes it 20.62-fold. The measured quantity is therefore not comparable across these four constructions, and the widest part of that incomparability involves the one that constrains composition most tightly.

The second is a boundary on the inversion this paper is named for. Across the three matched protocols a lower composition baseline goes with a *larger* contribution, which is the relation the

title describes. The shuffled arm has the lowest baseline of the four, by 0.1274, and the largest contribution, by +0.1860, so it falls on the same side of that relation rather than contradicting it.

But it does not extend the relation either, because shuffling is a different operation from matching. Matching makes negatives compositionally similar to positives in distribution, which raises the baseline; shuffling makes them identical pairwise, which removes the baseline. The two look alike, are described in similar language in method sections, and act on the measurement in opposite ways. This is the clearest case of the mechanism Section 3.4 identifies: what a negative-set protocol moves is the baseline, and the contribution follows.

No dataset was lost to the construction. Dinucleotide-preserving shuffling can return a near copy of a low-complexity window, which then has to be discarded as a negative; on this panel no positive was discarded either for a failed shuffle or for coming back more than 90% identical to its source.

3.6 The span holds for three model classes

Both neural models were trained on all three arms under the same folds, seed and code path, with per-window out-of-fold scores retained (Figure 4, Table 8).

Table 8: Nested contribution under three negative-set protocols for three model classes, $n = 94$ datasets, identical rows and folds. Spans are ratios of panel means with protein-clustered percentile intervals.

model	dinucleotide	GC	bias-aware	span
4-mer	+0.0663	+0.0265	+0.0122	5.42 (4.43–6.58)
CNN	+0.0837	+0.0330	+0.0113	7.42 (6.07–9.31)
SpliceBERT	+0.1735	+0.0890	+0.0466	3.72 (3.23–4.30)
composition alone	0.6274	0.7827	0.8248	

The span was present for every model class and was smallest, at 3.72-fold, for the largest model. The bias-aware protocol yielded the lowest contribution for all three model classes while carrying the highest composition baseline, so the inversion between apparent difficulty and measured contribution is not specific to a bag-of- k -mers measurement.

The apparent side of that inversion needs its own table, because until now it was reported for the 4-mer alone (Table 9). Dinucleotide matching is the hardest discrimination for all three model classes, by 0.11, 0.10 and 0.07 AUROC against GC matching. The bias-aware arm is the easiest for two of the three, and for SpliceBERT it is not: there GC matching is higher by 0.0039. So the ordering that makes the bias-aware protocol the easiest of the three is unambiguous on the composition baseline, where it leads by 0.042, and marginal on a strong model’s own AUROC. The inversion itself survives either reading, because the bias-aware arm yields the smallest contribution for every model while never being the hardest for any of them; what does not survive is the shorter statement that it is simply the easiest protocol.

Within a protocol the three model classes themselves span 2.62-fold (dinucleotide), 3.36-fold (GC) and 4.14-fold (bias-aware), which is the like-for-like comparison: the protocol effect is of

Table 9: Model-alone out-of-fold AUROC, three model classes under three protocols, $n = 94$ datasets, with the composition-only baseline for reference. Each model’s figure is pooled over every row that model covers, which is what a benchmark reports and which differs between models by 0.06% of rows. The rightmost column is the protocol each model finds easiest.

	dinucleotide	GC	bias-aware	easiest
composition alone	0.6274	0.7827	0.8248	bias-aware
4-mer	0.6879	0.7981	0.8169	bias-aware
CNN	0.7076	0.8084	0.8162	bias-aware
SpliceBERT	0.8099	0.8771	0.8732	GC

the same order as the effect of changing model class, not larger than it. The two-arm contrast was +0.0398 (CI +0.0325 to +0.0477) for the 4-mer, +0.0506 (CI +0.0417 to +0.0603) for the convolutional network and +0.0845 (CI +0.0757 to +0.0935) for SpliceBERT. The bias-aware arm also reorders two of the three model classes: the 4-mer’s +0.0122 there exceeds the convolutional network’s +0.0113, where the network is ahead of it in both other arms. The difference is +0.0010 with a protein-clustered interval of -0.0016 to $+0.0034$, so what the protocol does is not so much reverse the ranking as destroy it, leaving two model classes that the other two protocols separate confidently indistinguishable. Ranking methods is what a benchmark is for, so this is the most direct cost we can show. On the ratio scale the 4-mer and the CNN swap, with multipliers of 3.08, 3.35 and 2.33 (geometric mean over the 77 datasets positive in both arms for all three models), so the contrast does not increase along this ladder. The three classes differ in architecture, receptive field, pretraining and parameter count at once, so this is evidence that the effect is not specific to a bag-of- k -mers measurement, and not a controlled experiment on capacity.

Decomposing the log multiplier over the 262 dataset-by-model cells in which that model was positive in both arms, a looser rule than the 77-dataset intersection above, RBP identity accounted for 64.7% of variance, cell line for 0.1% and model class for 2.5%, with 33.3% residual. These shares are not directly comparable, and with type-II sums of squares they do not sum to one: a 79-level factor absorbs 29.7% of the variance of relabelled data whereas a three-level factor absorbs 0.8%. Against a stricter null that permutes protein labels between datasets while preserving dataset-by-model blocks, and which therefore absorbs 57.4% by itself, the excess attributable to protein identity was +7.3 points ($p = 0.003$).

That excess and the direct test below are one line of evidence and not two, because the protein factor is nearly the dataset factor at 79 levels against 94, so both rest on the fifteen proteins measured in both cell lines. The direct test is that the same protein’s log multiplier agreed between cell lines at $r = 0.582$ (Spearman 0.605) over 40 protein-by-model pairs. Those rows are not independent, since they come from fifteen proteins by three models sharing windows, labels and folds, so we report the protein-collapsed version as primary: averaging over models gives $r = 0.598$ ($p = 0.018$, Spearman 0.564, $p = 0.028$, $n = 15$), an order of magnitude weaker in p than the uncollapsed 0.0001 and the figure consistent with this paper’s own standard of resampling proteins. At fifteen proteins that point estimate carries very little: its Fisher z interval is +0.124 to +0.850, and a bootstrap

over the same fifteen proteins gives -0.012 to $+0.905$, which includes zero. We therefore read this as consistent with reproducibility across cell lines rather than as evidence for it, and the design has too few shared proteins to do better. Per model the agreement is $r = 0.754$ for the 4-mer and 0.766 for SpliceBERT but only 0.300 ($p = 0.34$) for the convolutional network, so it is not a property of all three. Cell line was not detectable, although with fifteen proteins measured in both lines the design has little power. Model class was small but detectable ($p = 0.023$), consistent with SpliceBERT’s lower multiplier, and robust to the censoring induced by requiring both arms positive.

We do not decompose the span into a compression term and a residual protocol effect. That decomposition requires transplanting a model’s discriminability increment across baselines, and its central assumption, that the increment is invariant to the baseline against which it is measured, is violated in these data. Two grids show it, and they are different grids. The first varies the transformation: six links, arcsine, complementary log-log, log, log-error, logit and probit, by the two transplant directions, twelve specifications in all. The residual is positive in every one of the twelve for all three models, from $+0.0127$ to $+0.1003$. The second varies the slope the transplant requires, over its three defensible sources, the GC arm’s, the dinucleotide arm’s and their mean, by the same two directions, so six specifications. There the residual’s bootstrap interval excludes zero in 4 of 6 for the 4-mer, 6 of 6 for the convolutional network and none of the 6 for SpliceBERT, and the point estimates span zero for two of the three models. The log-odds link reverses the sign entirely, which is the behaviour Mood [18] describes for odds ratios under unobserved heterogeneity. With three arms the residual’s sign follows the direction of transplant, $+0.0452$ carrying an increment from a high-baseline arm to a low-baseline one and -0.0259 in reverse. We report the raw contrast, which requires no transplant, link or transportability assumption.

All 564 score sets, across three arms and both neural model classes, are audited against the frozen fold assignment: every one is chromosome-grouped, with at most five chromosomes in a fold and no row having a same-strand neighbour within 1 kb assigned to a different fold.

That was not true of an earlier version of this analysis, and the history matters because the defect is of a kind that ordinary checks miss. For 20 of the 94 dinucleotide-arm datasets the neural scores had been produced under a stratified random partition. Fold sizes were preserved, so no count-based check detected it, while up to 23 chromosomes appeared in a single fold and up to 44.5% of rows sat within 1 kb of a same-strand neighbour in another fold. Those 200 fold-runs were retrained on the study partition and their scores replaced. The retraining lowered both neural models’ dinucleotide-arm contribution slightly, well inside their protein-clustered intervals; the 4-mer, which is refit on the study folds for every dataset, is unchanged, and the GC arm is unchanged to the last decimal place. The audit is `scripts/fold_integrity.py` and it runs inside the verification harness, so the property is asserted rather than remembered.

Two specification choices in the nested fit remain, and both are measured on the full panel rather than argued.

The first is the covariate scale. The 4-mer’s out-of-fold score is a log odds and the neural models’ are sigmoid probabilities, and logistic regression is not invariant to a nonlinear transform

of a covariate, so the model-class comparison put different scales in different arms. Putting the neural scores on the logit scale raises their contributions in every arm, by $+0.0008$ and $+0.0030$ in the GC arm, $+0.0004$ and $+0.0033$ in the dinucleotide arm and $+0.0004$ and $+0.0014$ in the bias-aware arm for the convolutional network and SpliceBERT respectively, with a maximum of 0.0219 on any single dataset. The two-arm contrast moves by -0.0003 and $+0.0003$. We retain the published scale because the effect is one-directional and raises the neural contributions, so every neural figure here is the conservative one, and because the contrast the paper is about moves in the fourth decimal against a protein-clustered half-width near 0.009. AUROC itself is invariant to monotone transforms, so no model-alone number is affected either way.

The second is the standardisation window. Columns are centred and scaled over the whole dataset before the out-of-fold loop, so a test row's own mean and standard deviation reach its features. This is improper, and it is also label-free and applied identically to every arm, so the argument that it cannot manufacture a contribution is sound but is still an argument. Standardising instead on each fold's training rows changes the panel-mean contribution by at most 0.00004 in any arm for any model, and by at most 0.0013 on any single dataset. That is below the fourth decimal the paper quotes, and it is the reason the whole-dataset form is retained rather than defended.

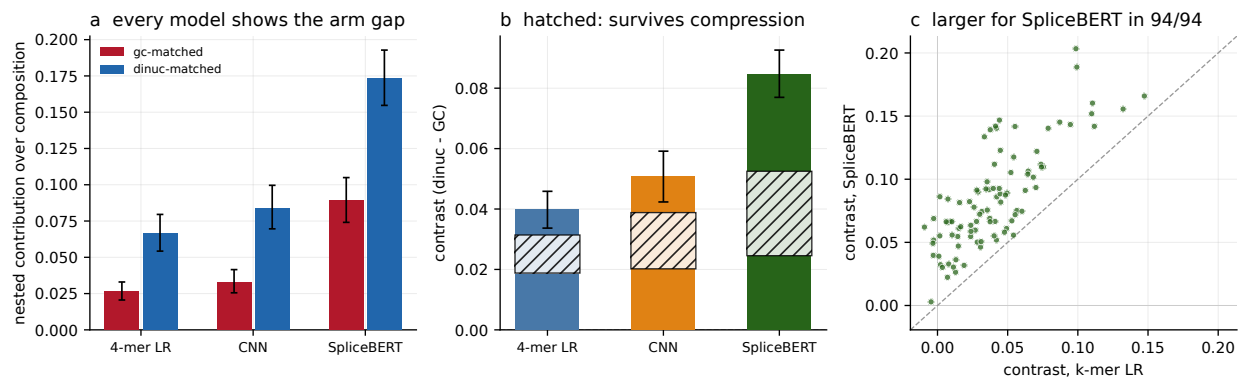


Figure 4: Three model classes on identical rows and folds, $n = 94$ datasets: a 4-mer logistic regression, a 7089-parameter convolutional network and a fully fine-tuned 19.78 M-parameter SpliceBERT. (a) Nested contribution in each composition-matched arm, per model; bars are 95% protein-clustered percentile intervals, as in panel (b). (b) The two-arm contrast per model, with 95% percentile intervals from a bootstrap over the 79 proteins taking all datasets of each sampled protein (4000 draws). The hatched band spans the protocol effect obtained by transplanting a discriminability increment in either direction, $+0.0188$ to $+0.0314$ for the 4-mer, $+0.0202$ to $+0.0388$ for the convolutional network and $+0.0246$ to $+0.0525$ for SpliceBERT: every member keeps the sign, so the contrast is not an artefact of AUROC compression, but its width is why we report the raw contrast and decline the decomposition in the text. (c) The contrast per dataset, SpliceBERT against the 4-mer; it is larger for SpliceBERT in all 94.

3.7 The protocol ordering does not depend on AUROC, but its magnitude does

Every quantity so far is an AUROC increment, and an AUROC increment is bounded above by $1 - c$ for a baseline c , which is the objection Section 3.4 spends its length answering. If the ordering is an artefact of that ceiling it should weaken or vanish on a scale with no ceiling. We therefore recomputed the same nested comparison on four further estimands, from the same two out-of-fold predictors so nothing is refitted (Table 10): the likelihood-ratio statistic $2(\ell_{\text{full}} - \ell_{\text{comp}})$, which is unbounded above; McFadden’s R^2 increment, which normalises it by the null deviance; the average-precision increment, which is prevalence-sensitive rather than rank-only; and the integrated discrimination improvement, a calibration-sensitive summary from the risk-prediction literature.

The ordering is the same on all five, for all three model classes: dinucleotide matching yields the most, GC matching next, the bias-aware protocol least, in 15 of 15 estimand-by-model cells. It survives on the deviance scale in particular, where no ceiling exists, so the direction of the protocol effect is not an artefact of measuring in AUROC.

The magnitude is another matter, and it is the honest cost of this exercise. The 4-mer’s three-protocol span is 5.42-fold in AUROC and 2.09-fold in deviance, 2.13 in McFadden, 3.70 in average precision and 2.40 in IDI; for the convolutional network the same figures run 7.42 down to 3.09, and for SpliceBERT 3.72 down to 2.06. The unbounded scales give the smaller spans in every case, which is what the ceiling objection predicts, so the five-fold figure this paper reports is a property of the AUROC scale and roughly a two-fold effect on scales without a ceiling. What is scale-free is the ordering, not the size.

Two choices in how the AUROC is aggregated across folds are open in the same way, and neither changes the ordering. Every AUROC here pools the five folds’ out-of-fold scores into one ranking; computing an AUROC per fold and averaging instead, or rank-normalising each fold’s scores to $[0, 1]$ before pooling, leaves dinucleotide $>$ GC $>$ bias-aware for all three model classes in all three aggregations. What the choice does affect is the level, and only for the neural models: pooling costs the convolutional network 0.0083 of contribution in the dinucleotide arm and SpliceBERT 0.0082, against 0.0002 for the 4-mer. The two alternatives agree with each other to within 0.0007, which is what one would expect if the cause is the five independently trained fold models not sharing a scale, since averaging within fold and rank-normalising within fold are two different repairs for exactly that. The 4-mer, whose fold predictors are linear scores from the same estimator, barely moves. Pooling is retained because it weights each row once rather than weighting a small fold as heavily as a large one, and the cost of retaining it is the figure above.

One diagnostic is not an increment and answers a sharper question. Regressing each model’s score on the composition block and taking the AUROC of the residual asks whether the part of the score orthogonal to composition discriminates at all. It does, in every arm: 0.63, 0.60 and 0.58 for the 4-mer under dinucleotide, GC and bias-aware negatives, 0.67, 0.66 and 0.63 for the convolutional network and 0.77, 0.76 and 0.72 for SpliceBERT. So a small increment is not the same as an absent signal, and the same ordering appears here too.

Table 10: The same nested comparison on five estimands, panel means over $n = 94$ datasets. The ordering dinucleotide $>$ GC $>$ bias-aware holds in all 15 estimand-by-model cells. Spans are the largest arm mean over the smallest. Deviance and McFadden are computed on the scale the model is fitted on and are unbounded above, so they carry no $1 - c$ ceiling.

estimand	model	dinucleotide	GC	bias-aware	span
AUROC increment	4-mer	0.0663	0.0265	0.0122	5.42
	CNN	0.0837	0.0330	0.0113	7.42
	SpliceBERT	0.1735	0.0890	0.0466	3.72
Δ deviance	4-mer	967	761	462	2.09
	CNN	1659	1185	536	3.09
	SpliceBERT	3524	2944	1656	2.13
McFadden R^2 increment	4-mer	0.0520	0.0380	0.0244	2.13
	CNN	0.0784	0.0523	0.0240	3.27
	SpliceBERT	0.1905	0.1543	0.0925	2.06
average precision increment	4-mer	0.0551	0.0276	0.0149	3.70
	CNN	0.0755	0.0355	0.0131	5.78
	SpliceBERT	0.1580	0.0917	0.0497	3.18
IDI	4-mer	0.0669	0.0448	0.0279	2.40
	CNN	0.1038	0.0666	0.0320	3.25
	SpliceBERT	0.2419	0.1851	0.1129	2.14

3.8 The span survives every order of composition baseline; the magnitudes do not, and the estimator stops working before order four

The composition baseline stops at order two because that is what a dinucleotide-preserving shuffle holds fixed. It is a defensible stopping point and it is a choice, so the same nested comparison was refit with the 63 trinucleotide frequencies added, on the full panel, in all three arms, for all three model classes, from the same out-of-fold model scores with nothing retrained (Table 11). At order two this recomputation returns spans of 5.42, 7.42 and 3.72, which are Table 8’s spans to four decimal places, so what follows is the paper’s own quantity measured one order up rather than a different quantity.

Table 11: Nested contribution over an order-three composition baseline, $n = 94$ datasets, identical rows, folds and model scores to Table 8. Surviving is the order-three contribution as a fraction of the order-two one, as a ratio of panel means; the range is over the three arms. Spans are ratios of panel means with protein-clustered percentile intervals.

model	dinucleotide	GC	bias-aware	span	surviving
4-mer	+0.0124	+0.0058	+0.0019	6.60 (4.71–10.05)	0.154–0.217
CNN	+0.0477	+0.0200	+0.0057	8.43 (6.49–11.64)	0.502–0.605
SpliceBERT	+0.1188	+0.0668	+0.0335	3.54 (3.05–4.11)	0.685–0.750
composition alone	0.6791	0.8023	0.8361		
rise from order two	+0.0516	+0.0196	+0.0113		

Every contribution shrinks, and by an amount that depends on the model class rather than on the protocol. The 4-mer retains between 0.154 and 0.217 of what it measured over an order-two baseline, depending on the arm; SpliceBERT retains between 0.685 and 0.750. For a bag of 4-mer counts this is close to tautological, since trinucleotide counts are linear aggregates of 4-mer counts and the only information a 4-mer model holds beyond a trinucleotide baseline is order four. The point is that most of what such a model is credited with contributing over a composition baseline is one further order of composition, not motif recognition and not any positional feature.

What does not shrink is the protocol dependence. The span across the three protocols is 6.60, 8.43 and 3.54 at order three against 5.42, 7.42 and 3.72 at order two, and the two-arm contrast remains +0.0067, +0.0277 and +0.0520 with every interval excluding zero. The direction of the movement is not uniform: for the 4-mer and the convolutional network the span grows, for SpliceBERT it falls slightly. Raising the baseline therefore removes much of the magnitude of what these benchmarks report and none of its protocol dependence, which is the same dissociation Section 3.7 found on the estimand axis.

Two features of the order-three numbers are cautions rather than results, and both are stated here on the full panel because the smaller panel misstated them in opposite directions. First, the spans of 6.60 and 8.43 have a near-zero denominator: the bias-aware arm’s 4-mer contribution is +0.0019 and is positive in only 60 of 94 datasets, against 71 of 94 for the convolutional network and 92 of 94 for SpliceBERT. The 4-mer span’s interval runs to 10.05 for that reason and the ratio should not be read as a magnitude; the absolute differences in Table 11 are the honest form. This is the same pathology that disqualifies normalising by the excess over chance in Section 3.4. Second, the surviving contribution is concentrated in a few datasets. On a 30-dataset subsample the three largest datasets held 51% of the GC arm’s positive mass; on the full panel they hold 21%. That fall is panel size and not dispersion, and reporting it alone would understate the caution: as a multiple of the share three datasets would hold if the mass were spread evenly, concentration is 4.66-fold on the full panel against 3.07-fold on the subsample. The mass is more concentrated here, not less.

Correcting for the headroom the raised baseline removes changes which model looks fragile, and does so inconsistently across arms. A nested increment is bounded above by $1 - c$, and the order-three baseline raises c , so part of every drop above is arithmetic. Transplanting each model’s order-two discriminability increment onto its order-three baseline and taking the residual, the 4-mer loses 1.29 times what SpliceBERT loses in the GC arm and 1.37 times in the dinucleotide arm, but 0.91 times in the bias-aware arm, where SpliceBERT loses marginally more. The reading that a k -mer model is the more fragile of the two under a raised baseline therefore holds in two of the three arms and reverses in the third, which is the same two-of-three pattern the apparent-difficulty ordering shows in Section 3.6.

Reporting two orders invites the question of why not a third, and answering it one order at a time invites it again, so we report the whole profile: orders one to four, all three arms, all three model classes, from the same out-of-fold scores (Table 12, Figure 5). Orders two and three reproduce the two published tables to machine precision, 10^{-16} , so the profile passes through both points

already in the paper rather than being a smooth curve near them.

Table 12: Nested contribution against the order of the composition baseline, $n = 94$ datasets. Baseline width counts columns after one is dropped per frequency family and entropy is appended. Order four is not a baseline: see the text.

	baseline width	order 1 4	order 2 19	order 3 82	order 4 337
dinucleotide	composition alone	0.5918	0.6274	0.6791	0.6861
	4-mer	+0.1010	+0.0663	+0.0124	+0.1444
	CNN	+0.1134	+0.0837	+0.0477	+0.0338
	SpliceBERT	+0.2106	+0.1735	+0.1188	+0.0930
GC	composition alone	0.7451	0.7827	0.8023	0.7961
	4-mer	+0.0616	+0.0265	+0.0058	+0.0986
	CNN	+0.0610	+0.0330	+0.0200	+0.0139
	SpliceBERT	+0.1267	+0.0890	+0.0668	+0.0548
bias-aware	composition alone	0.7959	0.8248	0.8361	0.8256
	4-mer	+0.0363	+0.0122	+0.0019	+0.0899
	CNN	+0.0285	+0.0113	+0.0057	+0.0036
	SpliceBERT	+0.0740	+0.0466	+0.0335	+0.0276

Over the three orders at which the baseline is a baseline, every contribution declines monotonically and the protocol dependence does not. The two-arm contrast is positive with an interval excluding zero in all twelve order-by-model cells, and the three-arm span runs from 2.78 to 6.60 for the 4-mer, 3.98 to 8.43 for the convolutional network and 2.85 to 3.72 for SpliceBERT. Where the baseline stops therefore sets the magnitude of what is reported and not whether the protocol matters.

Order four is a different object, and it is the most useful number in this section. A bag of 4-mer counts holds no information beyond an order-four composition block: at that order the baseline spans the model’s entire feature space, so the model’s true contribution is exactly zero by construction. The estimator does not report zero. It reports +0.0986, +0.1444 and +0.0899 in the three arms, positive on all 94 datasets in each, which is 3.72, 2.18 and 7.35 times the contribution the same estimator reports for the same model at order two. With the truth known, that is the instrument’s error, and it is larger than most of the increments this literature publishes.

The cause is measurable, and it explains why the effect is largest here rather than why it is absent elsewhere: Section 3.9 finds a smaller floor at order two, where the baseline is still improving. A 337-column baseline overfits at these sample sizes: the baseline’s own out-of-fold AUROC *falls* from order three to order four on 52, 34 and 64 of the 94 datasets, while from order one to order two it falls on 2, 3 and 1. A pre-fit model score is a one-column, well-regularised summary of the same information, so adding it repairs a fit the baseline could not achieve on its own, and the repair is credited as contribution. Two predictions follow and both hold: the effect should shrink as the sample grows, and it does, with Spearman correlations of -0.829 , -0.950 and -0.835 between dataset size and the floor; and it should appear only for the model whose features the block spans,

925 which is why the 4-mer’s contribution jumps at order four while the convolutional network’s and SpliceBERT’s continue to fall.

This qualifies the recommendation above. Raising the order of a composition baseline sharpens the comparison until the baseline itself stops fitting, and on panels of this size that happens before order four. A study raising its baseline should report whether the baseline’s own out-of-fold performance still improves, which costs nothing and is the diagnostic that separates a sharper instrument from a broken one. That diagnostic is necessary and not sufficient: the next section shows a floor at order two, where the baseline is still improving.

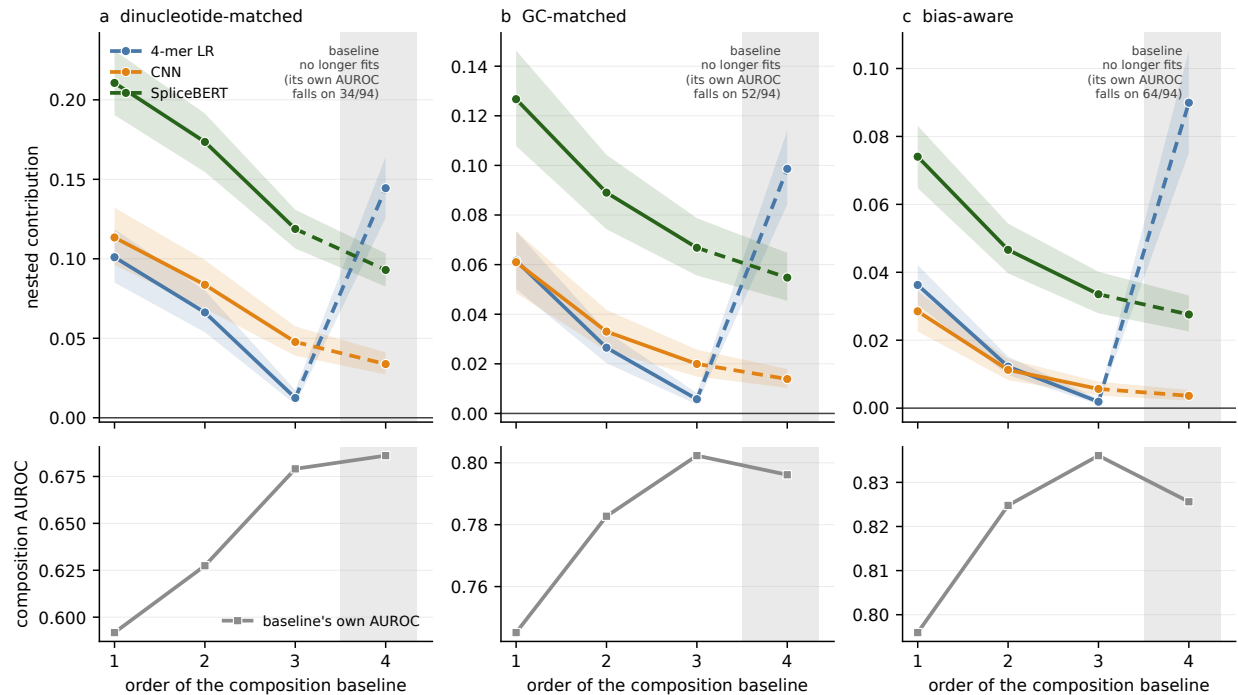


Figure 5: Nested contribution against the order of the composition baseline, $n = 94$ datasets, in each negative-set protocol: (a) dinucleotide-matched, (b) GC-matched, (c) bias-aware. Top row, contribution for each model class, with 95% protein-clustered percentile intervals as bands; the step from order three to order four is dashed because it crosses out of the range where the baseline is a baseline. Bottom row, the composition baseline’s own out-of-fold AUROC, which is the diagnostic: the shaded region marks order four, where a 337-column block overfits and its own AUROC falls on 34, 52 and 64 of the 94 datasets. The 4-mer’s contribution rises there while the two neural models’ continue to fall, because the order-four block spans the 4-mer’s feature space and its pre-fit score repairs a fit the baseline cannot achieve alone. Panel means can still rise where the per-dataset AUROC falls on a minority of datasets, which is why the counts are given.

3.9 The estimator has a positive floor where the true contribution is zero

935 Section 3.8 shows that a nested increment need not be zero when the truth is zero, but it shows it at a 337-column baseline, and every number reported here uses the nineteen-column one. The

same construction gives an exact null at the order we actually use. A 2-mer model scores a window by its sixteen dinucleotide counts; windows are a fixed 101 nt, so those counts are one hundred times the dinucleotide frequencies, and the fifteen frequency columns in the baseline together with the intercept span all sixteen exactly. A 2-mer’s score is therefore a linear function of features the baseline already contains, and its true nested contribution is zero by construction.

It does not measure zero. Fitting the 2-mer out of fold and passing its score through the same estimator gives +0.0119, +0.0137 and +0.0111 in the GC, dinucleotide and bias-aware arms (Table 13). The mechanism is the one Section 3.8 identifies: a supervised, out-of-fold projection of information the baseline already holds is a better conditioned summary of that information than the penalised nineteen-column fit achieves on its own, so adding it improves the fit without adding anything.

Two consequences follow and they point in opposite directions.

The span survives. A bias that is the same in every arm cannot produce a difference between arms, and this one very nearly is: the floor spans 1.24-fold across the three protocols while the contributions span 5.42-fold, so the floor covers 23% of the range it would have to explain. The protocol ordering is not manufactured by it.

The level does not survive, and one arm is affected severely. The floor is 44.7% of the GC arm’s reported contribution, 20.7% of the dinucleotide arm’s, and 90.4% of the bias-aware arm’s. The bias-aware arm’s +0.0122 is therefore not distinguishable from what this estimator returns on a model that contributes nothing, and every statement about that arm’s absolute contribution should be read as an upper bound. The arm’s position relative to the other two is unaffected, since the floor is nearly common to all three, but relative magnitudes are not: a floor-subtracted span would be far larger than the reported one, which is why we report the span without subtracting it. The 23% figure above is a ratio of fold-spans, so a floor with no across-arm variation at all would read 18% rather than zero; it bounds the contribution of the floor to the span conservatively rather than exactly.

Table 13: The nested estimator applied to a model whose information the baseline already contains, so the true contribution is zero. $n = 94$ datasets, order-two baseline, the same rows, folds and estimator as every other result here. Intervals are protein-clustered.

arm	measured floor	95% interval	reported contribution
dinucleotide	+0.0137	+0.0114 to +0.0163	+0.0663
GC	+0.0119	+0.0087 to +0.0155	+0.0265
bias-aware	+0.0111	+0.0085 to +0.0137	+0.0122

We report this as a property of the estimator rather than of our implementation of it. The quantity is the standard one in this literature: an increment in cross-validated AUROC from adding a model’s score to a baseline. Any study computing it on a supervised score inherits the same floor, and none of the seven sources in Table 6 reports a calibration that would reveal it.

3.10 One structural feature of the calibration replicates on an externally constructed benchmark

We applied the same decomposition to the negative sets released by Horlacher et al. [4], using their positives, peak calling, negative-set construction and fold assignments, over the 45 datasets shared with our panel (Table 14, Figure 6). The benchmark is external in its construction and not in its biology: the underlying experiments overlap ours, so this tests whether the same measurement behaves the same way when somebody else builds the negatives, not whether it behaves the same way on new proteins. Their released sets also carry no model-only AUROC, so difficulty there can only be read from the composition baseline, which is a weaker proxy than the one we use internally.

Table 14: The same decomposition applied to the released negative sets of Horlacher et al. [4], $n = 45$ datasets. The 4-mer’s own AUROC is not available for their negative sets, so difficulty on their benchmark is read from the composition baseline.

negative set	composition	composition + 4-mer	contribution
negative-1 (transcript background)	0.8211	0.8606	+0.0395
negative-2 (other RBPs’ sites)	0.7560	0.7726	+0.0166

The span on their benchmark was 2.381, against 2.17 for the corresponding family-matched pair in our own data, the GC-versus-bias-aware comparison. Their negative-2 is constructed like our bias-aware arm, which makes that the appropriate comparison and not our dinucleotide-versus-GC span of 2.50.

The two-family structure of the preceding subsection reproduced arm for arm. The within-arm relation between baseline and contribution was -0.645 ($p = 1.8 \times 10^{-6}$) for their composition-matched transcript-background arm and -0.187 ($p = 0.22$) for their other-RBPs’-sites arm, against -0.545 and -0.462 for our two composition-matched arms and -0.122 for our bias-aware arm. Two things did not replicate, and both belong here and not in the Limitations. The pooled relation across their two arms, the external counterpart of our -0.600 , is -0.202 ($p = 0.056$, $n = 90$), which is null at their sample size. And the direction of the level relation is reversed: their negative-2 has both the lower baseline and the lower contribution, so on their benchmark apparent difficulty and measured contribution move in the same direction rather than in opposite directions. A cross-family residual is consistent with that and a single-family headroom relation is not, but the ordering was not predicted in advance and we do not present it as a confirmation.

That second failure was previously read off the composition baseline, which is not the quantity the relation in the title is defined on, so we also measured the 4-mer’s own AUROC on their negative sets. It confirms the failure on the correct axis and sharpens it. Their negative-2 is harder for the 4-mer than their negative-1, 0.7647 against 0.8594 and harder in 41 of 45 datasets, and it also yields the smaller contribution, so the two move together. Per dataset the picture is not a reversed relation but no relation: the change in the model’s own AUROC and the change in contribution point in opposite directions in only 11 of 45 datasets, at Spearman -0.005 ($p = 0.97$). In our own data the

same two quantities point oppositely in 88 of 94 datasets for the GC-to-dinucleotide step and 57 of 94 for GC-to-bias-aware. So the inversion is a strong per-dataset regularity within this panel and is absent on the one benchmark we did not build.

One qualification on our own side, since the same statistic supplies it. Within our panel the inversion is a statement about DIRECTION and not about covarying magnitudes: the GC-to-dinucleotide step moves the two oppositely in 88 of 94 datasets, but across datasets the correlation between how much difficulty rose and how much contribution rose is $+0.385$ ($p = 1.3 \times 10^{-4}$), not negative. The datasets where a protocol costs the most apparent AUROC are not the datasets where it buys the most contribution, and nothing here claims they are.

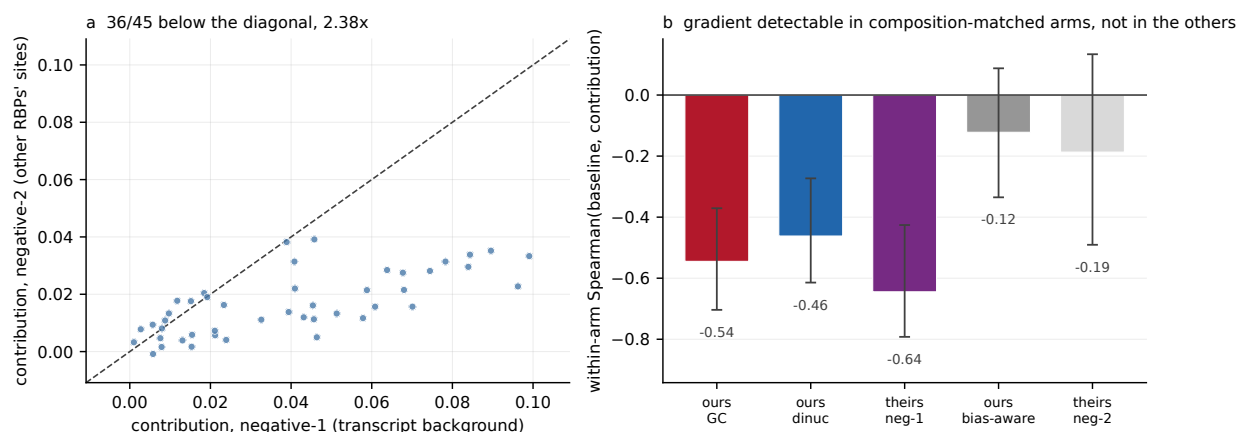


Figure 6: Replication using the released negative sets, peak calling and fold assignments of Horlacher et al. [4], over the 45 datasets shared with our panel; only the measurement is ours. (a) Nested contribution under their negative-2 against their negative-1, paired by dataset; 36 of 45 lie below the diagonal. (b) Within-arm Spearman correlation between the composition baseline and the contribution, for the composition-matched arms of both benchmarks and the other-RBPs'-sites arms of both, with percentile intervals over resampled proteins. The three composition-matched arms exclude zero and the two other-RBPs'-sites arms do not, independently in each benchmark. The grouping this invites is not stable to which term is placed on the horizontal axis and is not endorsed in Section 3.4; what the panel supports is the narrower statement about where the gradient is detectable.

3.11 Normalising by the baseline's headroom does not replicate out of sample

The recommendation this paper arrives at has two parts, and only one of them is testable. The first is to REPORT the composition-only AUROC obtained under the same protocol alongside every headline AUROC. That is a disclosure requirement rather than an empirical claim: it adds a number a reader can act on and removes none, so no experiment here could falsify it, and we do not present one. The second is to NORMALISE by that baseline's headroom, dividing the contribution by $1 - c$, in the hope of putting contributions from different protocols on a comparable scale. That is a substantive claim, we pre-specified two criteria that would falsify it, and this subsection is the test. It is the second part that fails, and the rest of this subsection is about the second part only.

1015 We tested it in the situation it is intended for, a reader holding two studies that used different protocols and wishing to compare what the models contributed (Figure 7).

On our data, normalising by headroom improved cross-protocol rank agreement in 3 of 3 protocol pairs and reduced disagreement on a common scale in 3 of 3, by 58%, 10% and 48%. Rank agreement is scale-free and cannot be flattered by normalisation. This held for all three model classes, with 1020 mean changes in rank agreement of +0.040 (4-mer), +0.094 (convolutional network) and +0.061 (SpliceBERT).

The improvement is weaker than it looks. Only the GC-versus-dinucleotide improvement has an interval excluding zero (+0.051, CI +0.007 to +0.115); the remaining two pairs are consistent in direction and individually null, and the pair that reaches significance is the one in which the 1025 protocol label contributes 0.05% of variance. That interval is marginal, not comfortable, and it does not survive adjustment for the three protocol pairs: the Bonferroni-corrected 98.33% interval is -0.003 to $+0.137$. The supporting sign evidence is weaker than it looks too, since three of three pairs has $p = 0.125$ under an exchangeable null, and the nine cells over three pairs by three model classes are not nine independent successes because they are drawn from the same 94 datasets and 1030 three arms. The test also fails to replicate out of sample, which matters more. On the 45 datasets of Horlacher et al. [4], rank agreement fell from 0.706 to 0.656 under normalisation and disagreement rose from 0.860 to 0.908, which are the two criteria we specified in advance as falsifying. The change in rank agreement was -0.050 (CI -0.222 to $+0.140$), so at $n = 45$ this is a failure to replicate rather than a refutation.

1035 The two parts therefore separate cleanly. We recommend reporting the composition-only AUROC obtained under the same protocol alongside every headline AUROC, on the grounds that it is the only summary of the protocol’s effect a reader can act on, and we note that this rests on that argument and not on a test. We do not recommend treating any normalisation, this one included, as rendering contributions comparable across protocols.

1040 4 Discussion

A negative-set protocol is a choice of question, not a setting on a difficulty dial. That is the single reading these results support, and it belongs before the numbers because the dial is the intuition they contradict. On a dial, a harder protocol is a more stringent test and a model that survives it has shown more. Here the protocol that made the discrimination hardest by AUROC produced the 1045 largest measured contribution, and the one that made it easiest produced the smallest, so difficulty and stringency came apart. The three protocols were not three settings of one control. They asked three different questions, and each answer is only interpretable against the question it answers.

How much a sequence model adds over sequence composition is therefore not a property of the model alone. Holding the model, the peak set, the folds and the estimator fixed, and varying only 1050 how the negative windows were assembled, the measured contribution of a 4-mer varied 5.4-fold, and between 3.7 and 7.4-fold across three model classes.

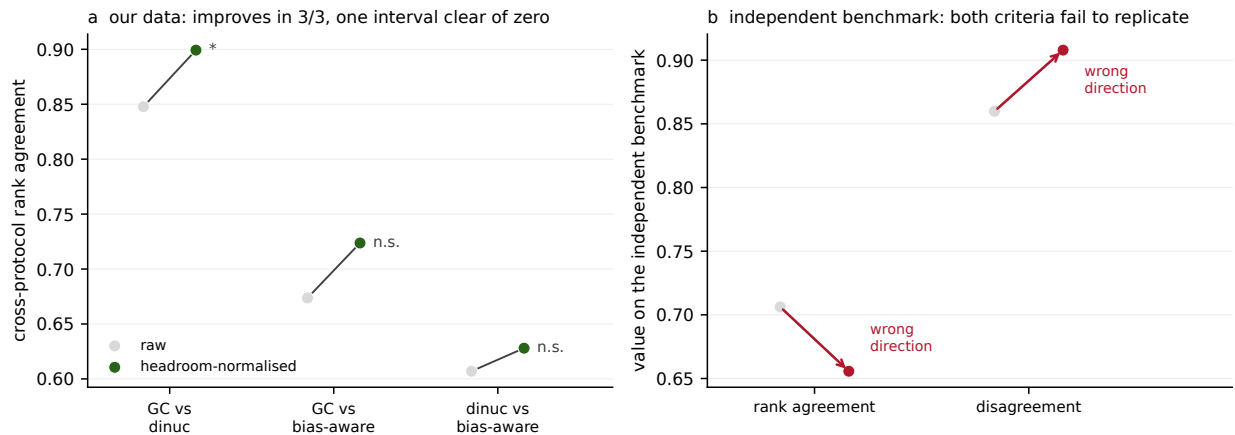


Figure 7: (a) Cross-protocol rank agreement between nested contributions before and after division by the headroom of the baseline, for each of the three protocol pairs in our data. Agreement improves in 3 of 3 pairs; only the GC-versus-dinucleotide improvement has a protein-clustered interval excluding zero. (b) The same test on the independent benchmark of Figure 6, $n = 45$: rank agreement falls and disagreement rises under normalisation, the two criteria specified in advance as falsifying. The change in rank agreement is -0.050 (-0.222 to $+0.140$), a failure to replicate rather than a refutation.

This has consequences for reporting practice, and they are specific enough to state as a checklist. A paper reporting a sequence model’s performance on RBP binding should give five things alongside its headline AUROC. The first three describe the measurement; the last two calibrate it.

1. **How the negatives were drawn**, in enough detail to rebuild the set: the candidate pool, the statistic matched on and its tolerance, the achieved match quality rather than the nominal one, and which covariates were left free. The two arms here differ in achieved GC matching in opposite directions at the median and in the tail; a nominal tolerance does not determine what was actually held fixed.
2. **The composition-only AUROC under that same protocol**, refit on those windows and those folds. It costs one logistic regression on nineteen features, and it is the only summary of the protocol’s effect a reader can act on. It is also frequently not small: on the bias-aware arm nineteen composition features beat the 4-mer outright in 53 of 94 datasets.
3. **The order of that baseline**, because the contribution is an increment over whatever the baseline captures and nothing fixes the stopping point. Ours stops at order two; at order three the span does not collapse, growing for two model classes and falling slightly for the third, while every contribution shrinks: by a quarter to a third for SpliceBERT and by roughly four fifths for the 4-mer. The baseline’s order sets the magnitude and is part of the measurement, not a detail of it.
4. **Whether the baseline’s own out-of-fold AUROC still improves** when its order is raised. This costs nothing, since the number is already computed, and it separates a sharper

instrument from a broken one: where the baseline has stopped fitting, the increment measured over it is estimator error rather than model contribution. On this panel the baseline still improves at order two and has stopped by order four.

1075 **5. A null increment for the estimator**, obtained by passing through it a model whose information the baseline already contains. Here a 2-mer serves, because its score lies in the span of the dinucleotide columns, and the estimator returns +0.0111 to +0.0137 rather than zero (Section 3.9). Any increment of that size is not distinguishable from what the estimator returns for nothing, and no study we surveyed reports such a calibration.

1080 Contributions measured under different negative-set protocols should not be compared. That withdraws a comparison instead of offering one, and it is the more consequential half, because cross-study comparison is what a benchmark is normally used for.

1085 Apparent difficulty is therefore not a proxy for the stringency of an evaluation. The bias-aware protocol is well motivated as a means of removing compositional and accessibility artefacts, and its design is not at fault; what does not follow is the inference from a lower or higher score to how much a model has learned. Separating one protein’s sites from another’s is largely a composition task, so a protocol that poses that task leaves a sequence model comparatively little to add. Two things make it a composition task, and only one of them is about binding sites. Distinct RBPs occupy distinct transcript regions, and the base composition characteristic of each region is readable from nineteen features; distinct RBPs also occupy compositionally distinct sites within a region. Rebuilding the bias-aware arm with the donor draw stratified on region removes the first and leaves the second: it costs the arm 0.020 AUROC of baseline, which is 46% of its excess over the GC arm, and a quarter of its measured contribution, and leaves its position among the three protocols unchanged. The lesson for a designer is that a protocol which frees a covariate the other protocols hold fixed is not simply a harder version of them, and that the comparison to make before attributing a baseline to biology is against the same protocol with that covariate restored.

1090 The relation between the composition baseline a protocol leaves and the contribution a model subsequently shows was strong for composition-matched negatives, both in our data and independently in that of Horlacher et al. [4], and not detectable for negatives drawn from the sites of other proteins. It is tempting to read a grouping off that result: two protocol families, divided by what kind of object the negative window is rather than by how tightly it is matched, with contributions comparable within a family and not between families. We do not think the data identify such a grouping, and say so because the reading is the natural one and it is the one our own figure invites.

1100 It fails because the grouping is not stable. The statistic correlates a difference with its own subtrahend, and depends on which term is placed on the horizontal axis, and under two of the three defensible choices the dinucleotide arm groups with the bias-aware arm instead of with the GC arm. The GC and dinucleotide protocols draw their negatives from the same kind of object, so a partition separating them cannot be a partition by kind of object. The only statement surviving every choice of axis is the weaker one that the GC arm is the outlier of the three. Two further qualifications point the same way: the pattern was recovered from the 4-mer and reproduced for neither neural

model, and on Horlacher et al. [4]’s benchmark the pooled relation is null. We therefore take from this subsection the gradient itself, for which the evidence is strong, and not a taxonomy of protocols built on top of it.

Protocol dependence can be reduced by fitting a transformation but not by choosing one from among the standard set. The smallest of the five that reduced it, division by the baseline’s headroom, is a normalisation rather than a transformation of the contribution, and is the pair of AUROCs we recommend reporting expressed as a single number. An exponent that closes the span appears to exist and does not survive inspection: it closes only under the one aggregation that a near-zero headroom can dominate, and under two aggregations that cannot it never approaches unity. Nor does it transport. The exponent fitted to our three protocols leaves an externally constructed benchmark essentially where it started; the exponent fitted to that benchmark is 2.4 times ours; and the exponent differs by a factor of 1.22 between model classes measured on the same data. This is the sense in which no protocol-free measure of contribution is available. The obstacle is not that no transformation works but that a transformation which works has to be refitted for each measurement, which is precisely what a comparable quantity cannot require.

Nor is it escapable by leaving the ROC. The same comparison on the likelihood-ratio statistic, McFadden’s R^2 increment, the average-precision increment and the integrated discrimination improvement gives the same ordering in all fifteen estimand-by-model cells, including on the deviance scale, which is unbounded above and so carries none of the $1 - c$ ceiling that a sceptical reading attributes the effect to. The estimand does change the size: the 4-mer’s three-protocol span is 5.42-fold in AUROC and about two-fold on the unbounded scales, and the unbounded scales are smaller in every case. So the direction of the protocol effect is a property of the comparison and the five-fold figure is a property of the AUROC scale, and this paper’s headline magnitude should be read as the latter.

Where the composition baseline stops is an analyst’s choice, and Section 3.8 measures what it costs. Its limiting case is Section 3.5: shuffling holds composition exactly, the baseline falls to a half, and the increment becomes the model’s own AUROC. Raising it to order three removed most of the 4-mer’s measured contribution and about a quarter to a third of SpliceBERT’s. For a bag-of- k -mers model most of what is reported as contribution over composition is one further order of composition rather than motif recognition or any positional feature. Any study reporting an increment over a composition baseline is reporting a quantity indexed by the order of that baseline, and few state which order was used. The protocol dependence survived the change and its magnitudes did not. A benchmark meant to separate model classes rather than rank them by how much short-range composition they absorb could raise the order of its baseline. That recommendation carries a circularity: trinucleotide counts are linear aggregates of 4-mer counts, so an order-three baseline is structurally unfavourable to k -mer models, and separates models by how much of their signal lies outside short-range composition, not by capacity as such.

Horlacher et al. [4] established that negative-set construction changes apparent performance across many RBP prediction methods and proposed the bias-aware alternative. The present work

1150 does not contest that and depends on it, since their released negative sets provide our only external validation. What we add is a calibration of it. What the protocol moves is not just apparent AUROC but the increment over an explicit baseline, and it moves that increment in the opposite direction, by a factor of between 3.7 and 7.4 depending on the model class, and no reparameterisation renders that increment comparable across benchmarks. On this panel the protocol effect is of the same order
 1155 as the effect of changing model class, which is what these benchmarks exist to resolve: within a protocol the three model classes span 2.62 to 4.14-fold, against 3.7 to 7.4-fold across protocols within a model. The same dependence has since been reported for transcription factor binding sites [5], on a different assay, a different class of model, and against a motif rather than a composition baseline, which suggests the problem is not specific to eCLIP. The circularity we cite in the Introduction for
 1160 variant-effect prediction has the same shape. Here it is measured for one estimand in one assay, with the arithmetic part separated from the part that is not.

Limitations

Every magnitude reported here is indexed to a mono- and dinucleotide baseline. The protocol dependence is not so indexed.

1165 The relation in the title holds throughout our own data, across three arms and three model classes, and does not hold on the one externally constructed benchmark we tested. On Horlacher et al. [4]’s two negative sets the arm that is harder for the 4-mer by its own AUROC also yields the smaller contribution, so difficulty and contribution move together there. Per dataset there is no relation at all rather than a reversed one, at Spearman -0.005 , against opposite movement in 88 of
 1170 94 datasets within our panel. Their two arms belong to different protocol families, which is the structure that would predict it, but that was not a prediction we made in advance.

Achieved matching is not exact matching. The GC matcher’s operative acceptance bound is 0.15 and not its nominal 0.05, holding the nominal tolerance for 94.8% of pairs, and the dinucleotide matcher reduces composition distance 2.27-fold, to a median L_1 of 0.220, so degrees-of-freedom
 1175 statements are exact for the design and approximate for the data.

All three model classes were measured under all three protocols, but the study covers two cell lines and one assay. Whether the calibration holds for other crosslinking protocols or other organisms is untested. The transform sweep, the baseline attribution, the recommendation test and the region-matched rebuild are reported for all three model classes. The order-three analysis is
 1180 reported for all three model classes at $n = 94$ on all three arms; where the earlier 30-dataset 4-mer subsample is quoted alongside it, the two use different bootstrap schemes, protein-clustered for the full panel and over datasets for the subsample. No model larger than 20 M parameters was tested, and no published RBP-specific method: the three classes are a capacity ladder, not a survey of the methods a benchmark would rank.

1185 Every model here is trained and evaluated under one protocol, so what we vary is the training set and the evaluation set together. A design that trained under one protocol and evaluated under another would separate what the evaluation set makes measurable from what the training set makes

learnable; Tourne et al. [5] performs that transfer and we do not. Our claims are therefore about the protocol as a whole choice, which is how a benchmark is used, and not about either half of it.

Three design parameters were fixed once and never varied, and nothing here speaks to their influence. The class ratio is 1:1 in every arm, one negative per positive, where a genome-wide screen faces a far smaller positive fraction. AUROC is invariant to prevalence, and the reported quantities are not biased by this. The nested fit is nonetheless estimated at a balance no application sees, and a precision-based estimand would move. The window is 101 nt throughout, and window length is precisely the resource the composition baseline spends. A longer window gives mononucleotide and dinucleotide frequencies more sequence to estimate from, which should raise the baseline and shrink the contribution in every arm, though not necessarily by the same amount in each. The third is the chromosome partition, which is frozen, and that one we measured rather than left open: rebuilding three alternative assignments under the same balanced-mass objective and the same minimum of three chromosomes per fold, and refitting the 4-mer on each, gives contrasts of +0.0392, +0.0396 and +0.0393 against +0.0398 on the partition we use. The range over the four is 0.00058, a thirteenth of the protein-clustered half-width. The choice of partition is not carrying the result. Our partition gives the largest of the four, by 0.00013 over the runner-up, so the headline sits at the top of that narrow range and not in the middle of it. Only the 4-mer can be re-partitioned this way: the neural sweeps trained on the frozen assignment; re-partitioning them would compare their old folds against new ones.

A related question is the grouping variable rather than the partition. Windows drawn from one gene are correlated, and a fold assignment splitting a gene across folds would leak. Chromosome grouping is strictly coarser than gene grouping and a locus lies on one chromosome, so that cannot happen here, and we verified it rather than asserting it: over both composition-matched arms, 22 of 825,718 gene groups fall in more than one fold, holding 176 windows. Every one is a gene *name* shared across chromosomes rather than a locus. Of the 72 such names in GENCODE v45, 36 are pseudo-autosomal chrX/chrY pairs and the remainder are multi-copy small-RNA families, the widest sharing one name across 24 chromosomes. Near-identical sequence at those loci does sit on both sides of a chromosome split. Family-level similarity is the one leakage channel this design does not close, at a scale of 176 windows in 2.7 million. Refitting the 4-mer under gene-clustered folds built to the same balanced-mass objective, which splits a median of 22.9 chromosomes across folds and is therefore the *less* conservative design, gives contributions of +0.0269 and +0.0664 against +0.0265 and +0.0663, and a two-arm contrast of +0.0394 against +0.0398. The finer grouping changes the contrast by 0.0003, so the coarser choice costs almost nothing and the grouping variable is not carrying the result.

The window centre is a fourth such parameter and the largest of them. Every window is centred on its peak's midpoint, peaks are not points, and Section 2.7 shows the discriminative signal sits a median 23 nt off that centre. The choice is demonstrably not aligned with the signal. The comparison usually asked for, summit-centred against midpoint-centred windows, cannot be made here: narrowPeak reserves column 10 for a point-source summit and it is -1 in every row of every

ENCODE eCLIP file in this panel: no summit exists to centre on. We therefore compared the three centres the interval does provide, on ten size-stratified datasets rebuilt from the genome in both composition-matched arms: the midpoint, the peak’s 5’ boundary in transcript orientation, which is where the crosslink sits, and the midpoint displaced 25 nt downstream, which has no rationale and is included as a bound on arbitrariness. The two-arm contrast is +0.0484, +0.0430 and +0.0407, a range of 0.0077. That is an order of magnitude larger than the chromosome partition’s 0.0006 or the fold grouping’s 0.0003, so of the design parameters left open the window centre is the one that matters most, and as with the partition the published choice gives the largest of the three rather than the middle.

Rebuilding the published centring also measures something we had not intended to measure. Both matchers sample their candidate windows, and a rebuild is a fresh draw of the same construction rather than the same negatives, and the spread between draws is the arm’s own run-to-run variability: at most 5.29×10^{-5} per dataset in the GC arm and 7.90×10^{-3} in the dinucleotide arm. The arm constraining fifteen degrees of freedom is the less stable of the two by two orders of magnitude, because matching sixteen frequencies well depends on which candidates the pool happened to contain, while a five-point GC tolerance is met by a large fraction of any pool.

The matcher itself has two free parameters, and varying them shows the same mechanism acting inside a single protocol. The dinucleotide matcher draws candidates at eight per positive and assigns each positive its nearest unused candidate greedily; both choices were made once. We rebuilt twelve size-stratified datasets under six settings, crossing greedy against an exact per-bucket assignment with three pool sizes. Assignment decomposes exactly by region and chromosome, since a candidate can only serve a positive from the same pool, so each bucket is solved to optimality. The pool floor had to be scaled alongside the multiple: at the published floor of 1500 candidates it binds in 92.3% of buckets; varying the multiple alone returns identical negatives and would have tested nothing.

Exact assignment improves the achieved match by 0.0036 to 0.0052 in mean L1, about 2%, and moves the measured contribution by at most 0.0020. The greedy approximation is defensible on the evidence rather than only on the argument given for it. Pool size does more: mean L1 falls from 0.2722 to 0.2259 between the smallest and largest pools. Across all six settings the contribution spans 0.0106 and the composition baseline spans 0.0446, and the two move in opposite directions at $r = -0.926$ ($p = 0.008$). An implementation detail with no name in any methods section therefore moves the reported quantity, and it moves it through the baseline, which is the same channel the protocol label acts through.

Rebuilding the published setting is a fresh draw rather than a reproduction, because the matcher samples its candidate pool and is deterministic only in the assignment given that pool. The resulting deviation from the published per-dataset gain has a median of 0.0031 and shrinks as the dataset grows, at Spearman -0.909 , which identifies it as sampling noise in the draw rather than drift in the construction.

One limitation concerns the provenance of the neural scores. Initialisation was unseeded across all 2830 committed fold-runs, at the cost quantified in Methods. A second, that 20 of the 94

dinucleotide-arm datasets carried scores from a partition that was not chromosome-grouped, has been removed rather than bounded: those 200 fold-runs were retrained on the study partition and all 564 score sets now pass the audit.

Two runs of the same code on different hardware bound what a reader reproducing this work should expect. The region-matched arm’s convolutional network was trained twice on one dataset, once on a Modal A10G and once on a single vCPU of a GCP Batch instance, with the same commit, the same 26,288 windows and the same five chromosome-grouped folds; the row sets, labels and fold assignments were identical on all five folds before anything was compared. Per-window scores correlate at Pearson 0.9903, lowest fold 0.9814, and the fold AUROCs differ by at most 0.0007. Initialisation is unseeded, so this bounds the device and the initialisation together rather than the device alone, which is also what a reader rerunning the released code would encounter. A practical consequence is worth recording: a 7089-parameter network runs only 1.65 times slower on one vCPU than on the accelerator, because a model that small never fills the device.

Positive sets differ slightly between arms. Window construction is identical, but a positive is dropped when its matcher finds no acceptable negative and the matchers fail on different windows, so the two composition-matched arms’ positive sets have a Jaccard similarity with median 0.9972 and minimum 0.9237 and are identical in 10 of 94 datasets. The negatives are therefore almost, but not exactly, the only difference between arms.

We considered restricting every arm to the intersection of its positive sets and declined. The intersection would make the negatives exactly the only difference, at the cost of discarding the windows on which the matchers disagree. Those are not a random sample: a matcher fails where a positive’s composition is hardest to match, so the intersection would preferentially drop the compositional extremes that the dinucleotide matcher works hardest on, and it would do so from the arm whose contribution is largest. Removing a discrepancy of under three parts in a thousand by deleting the observations most informative about the effect trades a small bias for a larger one, and the three-arm results in Section 3.6 are already computed on rows intersected across all three model classes, so the comparison that carries the paper’s headline is on identical rows regardless.

We applied no significance filter of our own to the positives, and none was available: ENCODE’s released eCLIP peak files are already thresholded, and across our 95 files the minimum log2 fold-enrichment is 3.0000 and the minimum $-\log_{10} p$ is 3.0003, matching the criteria of Van Nostrand et al. [3], with 91.7% of the 463,091 peaks carrying ENCODE’s IDR label and nine K562 datasets contributing non-IDR peaks. There were therefore no weak peaks to exclude. What we do discard is peak *width*: every positive is a fixed 101 nt window on the peak midpoint, so a 20 nt peak and a 2 kb peak are treated identically, and for the wider peaks the window may not contain the crosslink site. No expression filter was applied to negatives; that question is bounded by the control reported above and not removed. Region class is assigned from the window midpoint under a fixed precedence, with introns computed as gaps between exons and no gene-type filter, so a window spanning an annotation boundary can be labelled either way. This affects a matching covariate and not an outcome, and it is the same annotation for both classes in the two arms that match on it.

1305 The verification harness needs describing precisely, because its headline number invites more credit than it earns. Its 937 assertions compare each published value against a frozen expectations file generated from the same run those values come from, so they are regression coverage: they detect drift, corruption and silently skipped checks, and cannot confirm that the original run was correct. Two assertions do more, and they are the reproducibility claim worth making rather than
1310 the count: the 285 AUROCs recomputed from committed per-example scores, and the headline contrast recomputed from raw sequence. Each re-derives a value along a different path instead of comparing it against a record of itself.

Everything downstream of the neural scores is bit-reproducible; the scores themselves are not, because initialisation was unseeded, so a rerun gives slightly different scores and then identical
1315 numbers from every step after them. The harness asserts values, not the provenance of the code producing them, and the fold-partition defect above is exactly the class that distinction permits: it was found by inspection and by none of the 937 assertions, as was the region asymmetry of Section 3.2.

The manuscript audit traces every number here, decimals and bare counts alike, to a committed
1320 table, a configuration value or a constructed object, and no unsourced number remains. Three limits bound it. It cannot tell whether a number is attached to the right claim, so a real value on the wrong line passes. Its power is set by how densely sourced values fill the space it searches, about a quarter of both the four-decimal grid over $[0.5, 1.0]$ and the integer range the paper’s counts occupy, so a fabrication escapes roughly one time in four; the audit reports that occupancy on every
1325 run. And because it matches values against tables rather than claims against runs, it passed a stale statement of this section’s own coverage: the count above was quoted from an earlier run while a larger number of assertions were executing. That one is now checked against the live total on every run, which is the only class of number here that goes stale whenever the harness grows.

That density is also why AUROCs are quoted here to four decimal places, which is finer than
1330 the measurement warrants. A single dataset’s AUROC carries a DeLong standard error in the third decimal, so the fourth digit of any per-dataset value is not a claim about binding. It is a traceability device: the three-decimal grid over the range these values occupy is now about 88% full, so a three-decimal number matches some committed table almost by chance and the audit protects it hardly at all, while at four decimals the same check has real power. Panel-level quantities, which
1335 average over 94 datasets, do support the precision they are given.

Confidence intervals on a single dataset’s contribution are Wald intervals on the DeLong standard error and not bootstrap intervals, and the Firth [17] coefficient intervals reported in the accompanying repository resample rows, which is narrower than gene-level clustering would give.

5 Conclusion

1340 How much a sequence model adds over sequence composition is a joint property of the model and of the negative set it was evaluated against, and on this panel the two axes move against each other:

the protocol that made the discrimination easiest by AUROC was also the one that left a sequence model least to add.

The estimator that measures it is also not neutral. Given a model whose information the composition baseline already contains, so that the true increment is zero, it returns +0.0111 to +0.0137 on these data. That floor is nearly the same in every arm, so it does not create the difference between protocols, but it is nine tenths of the smallest arm’s reported value, so increments of that size say nothing about a model.

For a benchmark designer this has two operational consequences. Report the composition-only AUROC obtained under the same protocol alongside every headline AUROC; it costs one logistic regression on nineteen features and it is the only summary of a protocol’s effect that a reader can act on. And do not compare contributions measured under different negative-set protocols, nor read an increment as evidence of contribution without a null for the estimator that produced it. That restriction is one we can justify, where a normalisation that would remove the need for it is not. Of eight standard coordinates, five reduce the span without closing it, two are controls and one widens it. The exponent that appears to close it closes only under one way of aggregating the panel, and is in any case fitted rather than chosen, differing between benchmarks and between model classes measured on the same data.

A third consequence bears on how protocols are compared with each other. The three here hold different things fixed, and the one that frees transcript region pays for it in a baseline that a nineteen-feature model can read. Loosening a constraint does not make a protocol more stringent.

Data availability

All primary data are public. Positive windows derive from ENCODE eCLIP narrowPeak files [1, 3]; every dataset used, with its ENCFF file accession and ENCSR experiment accession, is listed in Supplementary Table S1 (95 datasets, 79 proteins, 95 experiments). Gene annotation is GENCODE v45 [9]. Cell-line expression used in the transcription control comes from four ENCODE polyA RNA-seq quantification files: ENCFF286KKZ and ENCFF829LCN for K562 (ENCSR115PIZ), and ENCFF863QWG and ENCFF376IXQ for HepG2 (ENCSR245ATJ and ENCSR813BDU). The external validation uses the negative sets and fold assignments released by Horlacher et al. [4] at [doi:10.5281/zenodo.10600977](https://doi.org/10.5281/zenodo.10600977), downloaded and md5-verified.

Derived data supporting every number in this paper, including per-window out-of-fold model scores for all three model classes and all per-dataset result tables, are released with the code under CC BY 4.0. Intermediate window tables containing genomic sequence are not redistributed; they are regenerated from the accessions above by the pipeline in one command.

Supplementary Table S1 lists all 95 panel datasets. One of them, NCBP2 in K562, did not clear the out-of-fold pair floor in the GC arm, so it carries no three-protocol results; its `in_three_arm_panel` field is false and its result columns are empty.

Code availability

All analysis code is available at <https://github.com/NirmalKumar31/rbp-protocol-calibration> under the MIT licence, with `results/` and `data/evidence/` under CC BY 4.0. Every published value is reproducible offline from the repository with `python scripts/verify.py --local results/tables`, which runs 937 numeric assertions against a frozen expectations file and asserts that no fewer than that number ran, so a check cannot silently skip. One quantity is an exception and is marked as such in the text: the design effect, whose block bootstrap needs the genomic coordinates that the published per-window score files do not carry. It is reproduced by `scripts/design_effect.py` against the window store, which is regenerated from the accessions above in one command.

Use of AI tools

Generative AI (Anthropic Claude) was used as a coding assistant to implement the analysis pipeline and to assist in drafting text, which the author revised. The study was conceived, designed and directed by the author, who made every analytical and interpretive decision, determined which claims the evidence supports, and is solely responsible for the content of this work. All reported values are produced by code in the accompanying repository and are reproduced by its verification harness; no reference was generated by a language model.

Acknowledgements

The author thanks Horlacher and colleagues for releasing their processed negative sets and fold assignments, without which the external validation in this paper would not have been possible.

Funding

This work received no specific funding. Cloud compute was paid for by the author.

Competing interests

The author declares no competing interests.

References

- [1] ENCODE Project Consortium. An integrated encyclopedia of DNA elements in the human genome. *Nature*, 489:57–74, 2012. [doi:10.1038/nature11247](https://doi.org/10.1038/nature11247)

- 1405 [2] E. L. Van Nostrand, G. A. Pratt, A. A. Shishkin, et al. Robust transcriptome-wide discovery of RNA-binding protein binding sites with enhanced CLIP (eCLIP). *Nature Methods*, 13:508–514, 2016. [doi:10.1038/nmeth.3810](https://doi.org/10.1038/nmeth.3810)
- [3] E. L. Van Nostrand, P. Freese, G. A. Pratt, et al. A large-scale binding and functional map of human RNA-binding proteins. *Nature*, 583:711–719, 2020. [doi:10.1038/s41586-020-2077-3](https://doi.org/10.1038/s41586-020-2077-3)
- 1410 [4] M. Horlacher, G. Cantini, J. Hesse, et al. A systematic benchmark of machine learning methods for protein–RNA interaction prediction. *Briefings in Bioinformatics*, 24(5):bbad307, 2023. [doi:10.1093/bib/bbad307](https://doi.org/10.1093/bib/bbad307)
- [5] N. Tourne, G. De Waele, V. Vermeirssen, and W. Waegeman. How negative sampling shapes the performance of transcription factor binding site prediction models. *Bioinformatics*, 42(2):btag048, 2026. [doi:10.1093/bioinformatics/btag048](https://doi.org/10.1093/bioinformatics/btag048)
- 1415 [6] A. Khan, R. Riudavets Puig, P. Boddie, and A. Mathelier. BiasAway: command-line and web server to generate nucleotide composition-matched DNA background sequences. *Bioinformatics*, 37(11):1607–1609, 2021. [doi:10.1093/bioinformatics/btaa928](https://doi.org/10.1093/bioinformatics/btaa928)
- [7] L.-M. Krützfeldt, M. Schubach, and M. Kircher. The impact of different negative training data on regulatory sequence predictions. *PLOS ONE*, 15(12):e0237412, 2020. [doi:10.1371/journal.pone.0237412](https://doi.org/10.1371/journal.pone.0237412)
- 1420 [8] E. Cohen-Davidi and I. Veksler-Lublinsky. Benchmarking the negatives: effect of negative data generation on the classification of miRNA-mRNA interactions. *PLOS Computational Biology*, 20(8):e1012385, 2024. [doi:10.1371/journal.pcbi.1012385](https://doi.org/10.1371/journal.pcbi.1012385)
- 1425 [9] A. Frankish, S. Carbonell-Sala, M. Diekhans, et al. GENCODE: reference annotation for the human and mouse genomes in 2023. *Nucleic Acids Research*, 51(D1):D942–D949, 2023. [doi:10.1093/nar/gkac1071](https://doi.org/10.1093/nar/gkac1071)
- [10] D. G. Grimm, C. A. Azencott, F. Aicheler, et al. The evaluation of tools used to predict the impact of missense variants is hindered by two types of circularity. *Human Mutation*, 36(5):513–523, 2015. [doi:10.1002/humu.22768](https://doi.org/10.1002/humu.22768)
- 1430 [11] S. Whalen, J. Schreiber, W. S. Noble, and K. S. Pollard. Navigating the pitfalls of applying machine learning in genomics. *Nature Reviews Genetics*, 23:169–181, 2022. [doi:10.1038/s41576-021-00434-9](https://doi.org/10.1038/s41576-021-00434-9)
- [12] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*, 44:837–845, 1988. [doi:10.2307/2531595](https://doi.org/10.2307/2531595)
- 1435 [13] X. Sun and W. Xu. Fast implementation of DeLong’s algorithm for comparing the areas under correlated receiver operating characteristic curves. *IEEE Signal Processing Letters*, 21(11):1389–1393, 2014. [doi:10.1109/LSP.2014.2337313](https://doi.org/10.1109/LSP.2014.2337313)

- 1440 [14] O. V. Demler, M. J. Pencina, and R. B. D’Agostino Sr. Misuse of DeLong test to compare AUCs for nested models. *Statistics in Medicine*, 31(23):2577–2587, 2012. [doi:10.1002/sim.5328](https://doi.org/10.1002/sim.5328)
- [15] M. J. Pencina, R. B. D’Agostino Sr, K. M. Pencina, A. C. J. W. Janssens, and P. Greenland. Interpreting incremental value of markers added to risk prediction models. *American Journal of Epidemiology*, 176(6):473–481, 2012. [doi:10.1093/aje/kws207](https://doi.org/10.1093/aje/kws207)
- 1445 [16] R. H. Somers. A new asymmetric measure of association for ordinal variables. *American Sociological Review*, 27(6):799–811, 1962. [doi:10.2307/2090408](https://doi.org/10.2307/2090408)
- [17] D. Firth. Bias reduction of maximum likelihood estimates. *Biometrika*, 80(1):27–38, 1993. [doi:10.1093/biomet/80.1.27](https://doi.org/10.1093/biomet/80.1.27)
- [18] C. Mood. Logistic regression: why we cannot do what we think we can do, and what we can
1450 do about it. *European Sociological Review*, 26(1):67–82, 2010. [doi:10.1093/esr/jcp006](https://doi.org/10.1093/esr/jcp006)
- [19] B. Alipanahi, A. DeLong, M. T. Weirauch, and B. J. Frey. Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. *Nature Biotechnology*, 33:831–838, 2015. [doi:10.1038/nbt.3300](https://doi.org/10.1038/nbt.3300)
- [20] D. Maticzka, S. J. Lange, F. Costa, and R. Backofen. GraphProt: modeling binding preferences
1455 of RNA-binding proteins. *Genome Biology*, 15:R17, 2014. [doi:10.1186/gb-2014-15-1-r17](https://doi.org/10.1186/gb-2014-15-1-r17)
- [21] K. Chen, Y. Zhou, M. Ding, et al. Self-supervised learning on millions of primary RNA sequences from 72 vertebrates improves sequence-based RNA splicing prediction. *Briefings in Bioinformatics*, 25(3):bbae163, 2024. [doi:10.1093/bib/bbae163](https://doi.org/10.1093/bib/bbae163)
- [22] C. R. Harris, K. J. Millman, S. J. van der Walt, et al. Array programming with NumPy. *Nature*,
1460 585:357–362, 2020. [doi:10.1038/s41586-020-2649-2](https://doi.org/10.1038/s41586-020-2649-2)
- [23] P. Virtanen, R. Gommers, T. E. Oliphant, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. [doi:10.1038/s41592-019-0686-2](https://doi.org/10.1038/s41592-019-0686-2)
- [24] F. Pedregosa, G. Varoquaux, A. Gramfort, et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- 1465 [25] A. Paszke, S. Gross, F. Massa, et al. PyTorch: an imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035, 2019.
- [26] J. D. Hunter. Matplotlib: a 2D graphics environment. *Computing in Science and Engineering*, 9(3):90–95, 2007. [doi:10.1109/MCSE.2007.55](https://doi.org/10.1109/MCSE.2007.55)